When Can Markets Crash Without News?

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Abstract. Dollar-neutral trading need not be price-neutral. We study this distinction in a deterministic multiasset market with fixed fundamentals, state-dependent price impact, paired-dollar transfers, price-level resilience, and zero terminal net executed dollars for each asset. A response-adapted price coordinate exposes the mechanism: paired trades preserve its sum, while resilience generally moves the system between the corresponding invariant surfaces. Terminal price displacements can therefore survive closed dollar ledgers, at a cost in share inventory. The resulting geometry yields global elasticity barriers, constructive reachability results, and execution-resource bounds. In an idealized ten-asset constant-impact witness, every asset finishes at half its fundamental price, with finite turnover and inventory but flow-completed instantaneous trades. Separately, an interval-certified finite-rate ten-asset witness under rolling price limits has seven assets falling more than 20.2 percent and an equal-weight geometric index falling more than 6 percent, with paired-dollar turnover below five. Arbitrarily severe partial-coverage losses remain constructively reachable with uniform finite resources for each fixed sublinear impact exponent, whereas full coverage faces an unattained horizon-dependent ceiling. These results distinguish structural impact barriers from the execution resources needed to realize systemic drawdowns.

Key words. market crashes; price impact; market resilience; systemic drawdown; reachability; execution constraints.

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1. Introduction.

Consider ten assets whose fundamental benchmark prices never change. Each executed sale must fund an equal-dollar purchase, and by the end of the horizon every asset's signed executed-dollar ledger must return to zero. Can all ten prices nevertheless close at half their initial values? Under a particular constant-impact, resilient-price specification the answer is yes. The construction requires substantial execution resources and idealized flow-completed instantaneous trades, but it is a fully dollar-closed price path. This apparent paradox motivates a precise question: which systemic terminal losses are *reachable*, which are excluded by market structure, and what resources does reaching them require? We study deterministic feasibility rather than crash likelihood, traders' equilibrium incentives, or a historical no-news event.

This reachability question connects two established lines of research. One examines price formation and execution, from Kyle's auction model [25] and Almgren–Chriss execution [5] to resilient order books [1, 3, 27] and large-trader or supply-curve models [6, 12]. Conditions excluding price manipulation, quasi-arbitrage, or inadmissible transient impact constrain the costs of trading programs [4, 17, 18, 21]. In no-manipulation analyses, a common benchmark is a *share-flat* round trip, which restores the share inventory; its cost properties depend on the specified impact and recovery laws. The second line studies price movements without contemporaneous public news through inference and hedging [19, 29], endogenous liquidity feedback [20], fire sales and portfolio contagion [11, 14, 15, 24], and predatory trading or

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cross-impact [2, 7, 9, 16, 30]. These studies illuminate trading costs and channels of price amplification. Our complementary object is the set of terminal price states attainable under *both* paired-dollar reallocation and assetwise dollar closure, without imposing a particular information, funding, or trading-motive mechanism.

These two accounting restrictions differ from the share-flat boundary condition. At each trade, *paired-dollar execution* requires $\sum_i \mathrm{d}X_i = 0$: purchases of one asset are funded by sales of another. At maturity, *assetwise dollar closure* additionally requires $X_i(T) = 0$ for every i. Neither condition implies $\theta_i(T) = \theta_i(0)$. Under the model's execution-cost identity, terminal price displacement and recovery dissipation are financed by consuming initial share inventory; dollar-flat reallocation is therefore distinct from a costless share-flat round trip. The distinction also depends on whether resilience acts on the price level or on displaced order-book volume [1, 28].

The organizing insight is geometric. Write $g(S) > 0$ for the effective price response per executed share. The response-adapted coordinate

$$\Phi(S) = \int_F^S \frac{u}{g(u)}\,\mathrm{d}u. \tag{1.1}$$

This coordinate satisfies $\mathrm{d}\Phi|_{\text{trade}} = \mathrm{d}X$. Consequently paired-dollar trades preserve $H(\boldsymbol{S}) = \sum_i \Phi(S_i)$, moving prices along its level sets. Price-level resilience generally changes H between trades. In particular, even if every cumulative dollar ledger closes at maturity, the price path can cross invariant surfaces during its intervening recovery episodes. Without resilience, the same terminal dollar closure would force every terminal price back to the benchmark. Figure 1 plots an explicit two-asset trade–recovery path and the ten-asset construction. The elasticity studied below is a local price-state response, distinct from cumulative metaorder impact scaling [10, 26].

This mechanism explains why the response shape matters. Let $e(S) = Sg'(S)/g(S)$ be the local impact elasticity. Under the paper's price-level recovery and positivity assumptions, $e(S) \geq 1$ on the relevant price range excludes a decline in *every* asset, whereas $e(S) \geq 2$ rules out positive losses under the combined average-log-loss and breadth criterion. At the other end, constant per-share impact permits arbitrarily severe losses over a proper subset of assets, provided a compensator is allowed; this extends to every fixed sublinear power exponent with severity-independent finite turnover and inventory bounds. Full asset coverage instead has a sharp, unattained time-dependent ceiling. These statements concern different terminal target sets and must not be conflated.

Two ten-asset benchmark families make the distinction tangible. In the idealized full-coverage construction (A1), $N = 10$, $F = \eta = 1$, $\rho = 0.4$, and $T = 40$ yield $S_i(T) = 0.5$ for all ten assets, an ordinary price fall of 50% per asset ($\log 2$ in log-loss units), with paired turnover below 35 and maximum cumulative share sales below 17 per asset. An idealized nine-target construction (A2) further illustrates unbounded partial-coverage log severity with a uniform finite resource upper bound. By contrast, a distinct, absolutely continuous and interval-certified rolling-band witness (B1), with $N = 10$, $F = \eta = 1$, $\rho = 0.4$, $T = 40$, and uniform two-session $\pm 25\%$ rolling limits, produces ordinary price declines above 20.2% in seven assets and a geometric-index fall above 6%, with gross share rates below 0.285, per-asset cumulative share sales below two, and paired turnover $M < 5$. With a larger turnover allowance ($M < 7$), a

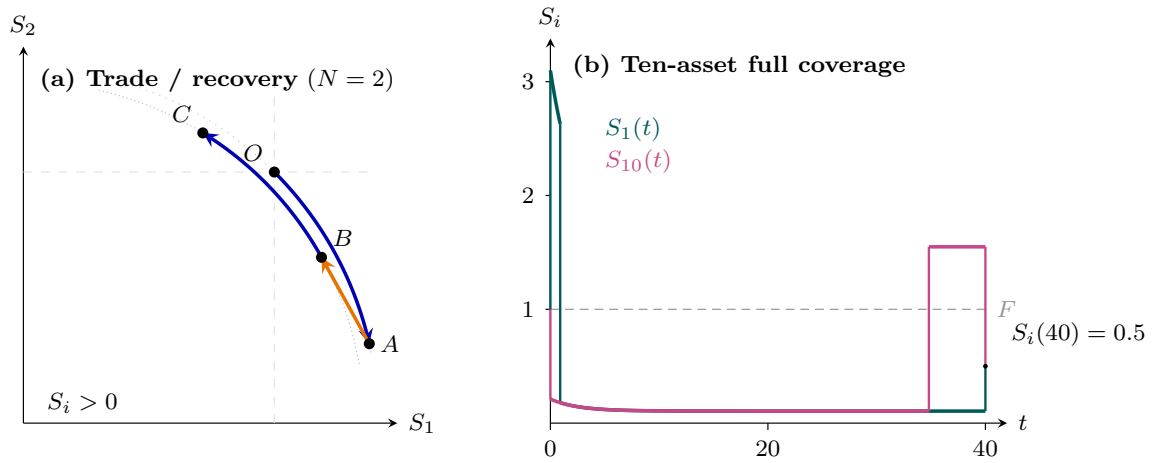


Figure 1. ***Conservation geometry and resilience-driven reachability.*** *(a) Under constant impact $g = 1$ with $N = 2$ and $F = \rho = 1$, equal-and-opposite paired-dollar pulses of magnitude 0.45 follow the distinct fixed-radius curves $O \to A$ and $B \to C$; recovery over $T = 0.7$ moves the state from A to B and changes the conserved radius. Both individual executed-dollar ledgers close, yet $C \neq O$ and the mean log loss is positive. (b) Selected prices in the analytical ten-asset full-coverage witness with $\rho = 0.4$ and $T = 40$; the other eight assets take the same cyclic high and low roles. Flow-completed pulses produce the jumps. All ten prices finish at $F/2$ with $X_i(40) = 0$. Both panels maintain strictly positive prices along trajectories evaluated from the model's analytic primitives.*

companion certified path (B2) achieves declines above 2% across all ten assets. Together, the benchmarks contrast idealized radial reachability with rate- and band-constrained realizations.

Our contributions are connected stages of a single argument. First, response-adapted conservation geometry and its interaction with resilience yield *structural barriers* and perturbation bounds for imperfect dollar pairing or heterogeneous impact. Second, power-law constructions and the sharp full-coverage boundary distinguish *reachable from excluded terminal loss states*. Third, turnover, inventory, bounded-rate and rolling-band analysis establish *execution-resource requirements* and certified constrained realizations. Section 2 derives the coordinates and accounting protocol; Section 3 develops the barriers; Section 4 proves the reachability results with the idealized ten-asset examples; Sections 5–6 treat resources and rates; Section 7 gives constrained certificates; and Section 8 discusses the economic meaning and limits. The appendices preserve the full constructive proofs and certification methods.

2. Model and Systemic Drawdown Criterion.

We specify a risky-asset reallocation submarket in which every executed sale is matched by an equal-dollar purchase. Settlement and price response are distinct assumptions: the former constrains flows, while the latter determines the induced price motion. All subsequent results use Assumption 2.1 unless stated otherwise.

Assumption 2.1 (The resilient paired-dollar market). Fix $N \geq 2$ assets and a finite horizon $[0, T]$. The following jointly define the admissible market.

(M1) **Positive prices and coordinates.** Each transaction price $S_i(t)$ is strictly positive and initially $S_i(0) = F > 0$. A strictly increasing C^1 coordinate $q_i = \phi_i(S_i)$ preserves

price priority. The analysis concerns deterministic admissible execution paths.

(M2) **Trade and execution accounting.** Executed share change is the signed quantity $\mathrm{d}\theta_i$; unsigned volume counts its absolute increments, not both counterparties twice. Executions obey marginal-price settlement $\mathrm{d}X_i = S_i\,\mathrm{d}\theta_i$. Matching uses price priority conditional on an order book; the effective price response is a separately imposed, positive, locally C^2 function $g(S)$, such that $\mathrm{d}S_i|_{\mathrm{trade}} = g(S_i)\,\mathrm{d}\theta_i$.

(M3) **No fundamental news and resilience.** The benchmark F stays fixed. Between trades, prices recover toward F at a common constant rate $\rho > 0$: $\dot{S}_i = -\rho(S_i - F)$ in the absence of execution. The response function g and ρ are specified exogenously.

(M4) **Paired-dollar transaction network.** Each permitted transaction moves an equal executed-dollar amount between a source asset and a destination asset: $\mathrm{d}\boldsymbol{X} = B\,\mathrm{d}\boldsymbol{z}$, where B is the oriented incidence matrix of the complete bidirectional graph, $\mathrm{d}\boldsymbol{z} \geq 0$, and $\mathbf{1}^\top B = 0$. This condition specifies an idealized paired-reallocation submarket without external cash inflows.

(M5) **Admissible controls and terminal closure.** Paths are piecewise continuous and of finite execution variation, with finitely many marginally settled instantaneous pulses allowed; every price remains positive and required sale inventory is available without short sales. The terminal comparison imposes $X_i(T) = 0$ for *each* asset, not $\theta_i(T) = \theta_i(0)$. Where resource caps are claimed, identical total paired-dollar turnover and per-asset starting share limits are imposed across markets.

Price-level resilience as a modeling choice. Assumption (M3) fixes recovery in the *price* coordinate, $\dot{S} = -\rho(S-F)$, separately from the effective trade response in (M2). Recovery in the order-book volume coordinate is a different model, except for constant impact; Subsection 8.2 examines which results survive that replacement. The structural elasticity barriers below use (M3).

Two levels of executed-dollar accounting. Condition (M4) matches dollars within each transfer, while (M5) closes each asset's cumulative signed-dollar ledger at T; the associated inventory accounting follows from (2.11).

Finite-variation paths and marginally settled pulses. For absolutely continuous controls the differential notation is classical. At a jump, settlement follows the within-event price path rather than valuing the entire transaction at an endpoint. Hold physical time fixed and parameterize a directed edge $e : i \to j$ carrying dollar mass $z > 0$ by $a \in [0, z]$:

$$\frac{\mathrm{d}\widetilde{S}_i}{\mathrm{d}a} = -\frac{g(\widetilde{S}_i)}{\widetilde{S}_i}, \quad \frac{\mathrm{d}\widetilde{S}_j}{\mathrm{d}a} = \frac{g(\widetilde{S}_j)}{\widetilde{S}_j}, \quad \frac{\mathrm{d}\widetilde{\theta}_i}{\mathrm{d}a} = -\frac{1}{\widetilde{S}_i}, \quad \frac{\mathrm{d}\widetilde{\theta}_j}{\mathrm{d}a} = \frac{1}{\widetilde{S}_j}. \tag{2.1}$$

The resulting endpoint flows satisfy $\Delta X_i = -z$, $\Delta X_j = z$ and $\Delta\Phi(S_i) = -z$, $\Delta\Phi(S_j) = z$. Equivalently, a single-share impulse is evaluated by solving $\mathrm{d}\widetilde{S}/\,\mathrm{d}\xi = g(\widetilde{S})$ and integrating $\int \widetilde{S}\,\mathrm{d}\xi$ along that response trajectory. These *flow-completed* (Marcus-type) updates preserve the ordinary chain rule; graph-completed solutions of bounded-variation impulsive control systems provide a related analytic framework [8]. Finite pulses on distinct edges are processed in their prescribed order. A pulse is admissible only while all intermediate prices remain strictly positive. These rules specify the interpretation of all jump expressions in what follows.

From the conditional matching map to an effective response. Fix a frozen book $\mathcal{B}$ with locally positive continuous share-depth density $\rho_S(S;\mathcal{B})$. A one-sided sweep whose signed marginal share size is $\mathrm{d}\theta$ obeys the local relation

$$\mathrm{d}\theta = \rho_S(S;\mathcal{B})\,\mathrm{d}S, \qquad \rho_q(q;\mathcal{B})\,\mathrm{d}q = \rho_S(S;\mathcal{B})\,\mathrm{d}S. \tag{2.2}$$

The exact discrete version uses the generalized inverse of the book's cumulative depth. In the continuous approximation, changing $q = \phi(S)$ changes the density numerically through the Jacobian, not the matching priority. The relation concerns the marginal executed price; dynamic midpoint evolution, replenishment, and cross-impact require additional dynamics. In large-trader and supply-curve models, execution prices likewise respond to trading activity [6, 12]. Assumption (M2) uses the frozen-book relation to motivate the effective state-dependent response $g(S) = 1/\rho_S^{\mathrm{eff}}(S)$.

Price dynamics for N assets. Under the full market specification, the level-price law is

$$\mathrm{d}S_i(t) = -\rho\big(S_i(t) - F\big)\,\mathrm{d}t + g(S_i(t))\,\mathrm{d}\theta_i(t), \qquad \mathrm{d}X_i = S_i\,\mathrm{d}\theta_i. \tag{2.3}$$

Choose the strictly monotone generalized coordinate $q_i = \log(S_i/F)$. Conversion to paired dollars gives

$$\mathrm{d}q_i\big|_{\mathrm{trade}} = \frac{g(S_i)}{S_i^2}\,\mathrm{d}X_i. \tag{2.4}$$

Combining this conversion, transaction pairing, and resilience yields the full multiasset system

$$\mathrm{d}\boldsymbol{q} = -\rho(\mathbf{1} - e^{-\boldsymbol{q}})\,\mathrm{d}t + D(\boldsymbol{q})B\,\mathrm{d}\boldsymbol{z}, \qquad D_{ii}(\boldsymbol{q}) = \frac{g(Fe^{q_i})}{F^2 e^{2q_i}}. \tag{2.5}$$

The exponential acts componentwise. This log coordinate is convenient for measuring losses, but executed dollars admit a simpler *response-adapted* coordinate. Let $y_i = \Phi(S_i)$ with Φ as in (1.1). It is strictly increasing and satisfies $\Phi(F) = 0$. Moreover, among differentiable coordinates ψ normalized by $\psi(F) = 0$, the requirement that each marginal trade obey $\mathrm{d}\psi(S)|_{\mathrm{trade}} = \mathrm{d}X$ forces $\psi'(S)g(S) = S$ and thus uniquely selects $\psi = \Phi$. Define the restoring response

$$\mathcal{R}(y) = \frac{S(S-F)}{g(S)}, \qquad S = \Phi^{-1}(y).$$

The entire multiasset market then has the *trade-normalized form*

$$\mathrm{d}\boldsymbol{y} = B\,\mathrm{d}\boldsymbol{z} - \rho\mathcal{R}(\boldsymbol{y})\,\mathrm{d}t, \qquad \mathbf{1}^\top B = 0, \tag{2.6}$$

where $\mathcal{R}$ acts componentwise and the execution rule applies also to flow-completed pulses. Trading is therefore an additive zero-sum input; all nonlinear price-state dependence has moved into the restoring term and the inverse map $S = \Phi^{-1}(y)$. The transformed state space is $\mathcal{J}^N$, where $\mathcal{J} = \Phi((0,\infty))$. For $g_p(S) = \eta(S/F)^p$, the range is $(-F^2/[\eta(2-p)],\infty)$ for $p < 2$, $\mathbb{R}$ for $p = 2$, and $(-\infty, F^2/[\eta(p-2)])$ for $p > 2$. Thus additive trading inputs do not remove strict price positivity or execution constraints. This change of coordinates does not alter which price states are reachable. With $\rho = 0$ the traded sum of generalized coordinates is exactly conserved.

As illustrated in Figure 1, trading moves the state along level sets of $H = \sum_i \Phi(S_i)$, while recovery moves it across them.

Definition 2.2 (Terminal systemic drawdown). *Let* $L_i(T) = \log[F/S_i(T)]$ *and* $\overline{L}(T) = N^{-1}\sum_i L_i(T)$. *The* k*th-largest loss, where* $k = \lceil \alpha N \rceil$, *is* $Q_\alpha(T)$. *Define*

$$\mathcal{C}_\alpha(T) = \min\{\overline{L}(T), Q_\alpha(T)\}. \tag{2.7}$$

For $\delta > 0$, *a* terminal crisis *occurs when* $\mathcal{C}_\alpha(T) \geq \delta$. *Thus the mean loss and at least* k *individual losses must each reach* δ. *The target is selected independently of the price-response function.*

The mean-loss component measures the decline of the equally weighted geometric price index, since

$$\overline{L}(T) = -\log\left[\left(\prod_{i=1}^{N}\frac{S_i(T)}{F}\right)^{1/N}\right].$$

The criterion therefore combines geometric-index deterioration with a breadth requirement. Arithmetic price averages and market-capitalization-weighted indices are different aggregates; for example, the $p = 1$ market below can have $\mathcal{C}_\alpha(T) > 0$ while $\sum_i S_i(T) = NF$.

The discrete coverage rule is material: for $N = 3$, $\alpha = 0.9$ means $k = 3$, so it requires a fall in all three assets. More generally, the contrast $k = N$ versus $k < N$ determines whether a compensating asset exists at all. This same distinction appears first in the A/B/C classification and later in the multiday cross-asset capacity inequality. A first-passage event $\tau_{\alpha,\delta} = \inf\{t : \mathcal{C}_\alpha(t) \geq \delta\}$ is different from a *terminal* crisis under dollar closure; we study the latter.

Definition 2.3 (Turnover, share inventory and terminal critical budget). *For an admissible path* γ, *set*

$$M(\gamma) = \sum_e \int_0^T \mathrm{d}z_e, \qquad H_i^-(\gamma) = \int_0^T \big(-\,\mathrm{d}\theta_i\big)_+. \tag{2.8}$$

Under allowed initial inventory H_i *and terminal dollar closure, define*

$$M_*^{\mathrm{end}} = \inf\{M(\gamma) : X_i(T) = 0\ \forall i,\ \mathcal{C}_\alpha(T) \geq \delta,\ \gamma\ \textit{admissible}\}, \tag{2.9}$$

with $\inf \varnothing = +\infty$. *A paired dollar transfer is counted once in* M, *although it appears with opposite signs in two asset ledgers.*

Execution-cost identity. Define

$$E(S) = \int_F^S \frac{u - F}{g(u)}\,\mathrm{d}u \geq 0. \tag{2.10}$$

For any admissible path beginning at F the execution-cost identity is

$$\int_0^T S_i\,\mathrm{d}\theta_i = F[\theta_i(T) - \theta_i(0)] + E(S_i(T)) + \rho\int_0^T \frac{(S_i - F)^2}{g(S_i)}\,\mathrm{d}t. \tag{2.11}$$

Thus a *share-flat* round trip has nonnegative executed cost under (M2)–(M3). The paper's terminal condition $X_i(T) = 0$ instead closes executed *dollars*, not shares: a terminally dollar-flat path may consume initial inventory. Appendix A proves the accounting identity and its consequences for admissible controls.

Comparison with share-flat trading. The terminal dollar condition and the standard share-flat condition are not interchangeable. For example, with $g \equiv \eta$, a single asset purchased by a sufficiently small $a > 0$ shares at time 0 and sold by a shares at time T has $\theta(T) = \theta(0)$ but terminal price $S(T) = F - \eta a(1 - e^{-\rho T}) < F$; its net executed dollars are not zero and the positive cost in (2.11) is paid through the cash account. Conversely, under (M4), requiring every asset to be share-flat *as well as* terminally dollar-flat would force $\sum_i [E(S_i(T)) + \rho \int_0^T (S_i - F)^2/g(S_i)\, \mathrm{d}t] = 0$. Thus there can be no positive-time dissipative price excursion under both closures together. The dollar-flat protocol deliberately isolates an economic question distinct from standard share-round-trip execution.

3. Structural Barriers and Robustness.

The normal form (2.6) supplies a conserved scalar during trading. Write $H(\boldsymbol{S}) = \mathbf{1}^\top \boldsymbol{y} = \sum_i \Phi(S_i)$. Summing over assets cancels every matched dollar transfer and leaves only recovery:

$$\mathrm{d}H = -\rho \sum_{i=1}^{N} \mathcal{R}(y_i)\, \mathrm{d}t = -\rho \sum_{i=1}^{N} \frac{S_i(S_i - F)}{g(S_i)}\, \mathrm{d}t. \tag{3.1}$$

The scalar H is invariant under trading and evolves through price recovery between trades. Its restoring drift need not have a single sign along an arbitrary path: reallocation changes the configuration at which the restoring field acts. The level surfaces alone therefore neither prove a barrier nor guarantee a crash; the relevant question is whether the coupled trading and recovery dynamics can enter the terminal crisis region. Curvature comparisons provide the answer for the structural exclusions below. Introduce the response elasticity

$$e(S) = \frac{S g'(S)}{g(S)}. \tag{3.2}$$

The derivatives of Φ and $G(x) = \Phi(Fe^x)$ are

$$\Phi''(S) = \frac{1 - e(S)}{g(S)}, \qquad G''(x) = \frac{S^2}{g(S)}\,[2 - e(S)], \quad S = Fe^x. \tag{3.3}$$

The distinction between these two curvatures tracks the definition of systemic loss. A barrier to *every asset* being below its benchmark is weaker than a barrier to our functional, which also involves the *average* log loss. Thus the same coordinate can produce two different exclusion statements.

Theorem 3.1 (Barrier to a decline of every asset). *If $e(S) \geq 1$ at all prices attainable by admissible paths, then $S_i(t) < F$ for every i is impossible at any time t. In particular, no positive crisis with $k = N$ is reachable.*

Proof. Equation (3.3) makes Φ concave. Its tangent inequality at S is $\Phi(F) \leq \Phi(S) + \Phi'(S)(F - S)$, hence $(S - F)\Phi'(S) \leq \Phi(S)$. Consequently (3.1) implies $\dot{H} \geq -\rho H$ between trades; pulses preserve H. Since $H(0) = 0$, comparison gives $H(t) \geq 0$. If every $S_i(t) < F$, strict monotonicity and $\Phi(F) = 0$ instead imply $H(t) < 0$, a contradiction. ∎

Theorem 3.2 (Barrier to a systemic crisis of any breadth). *If $e(S) \geq 2$ on the admissible price range, then for every admissible control, $N \geq 2$, $\alpha \in (0, 1]$ and $t \in [0, T]$,*

$$\mathcal{C}_\alpha(t) \leq 0. \tag{3.4}$$

Thus no strictly positive systemic terminal-crisis threshold can be reached.

Proof. The preceding theorem gives $H(t) \geq 0$. The second identity in (3.3) makes $G(x) = \Phi(Fe^x)$ concave and strictly increasing. Jensen's inequality yields

$$0 \leq \frac{H(t)}{N} = \frac{1}{N}\sum_i G(q_i(t)) \leq G\left(\frac{1}{N}\sum_i q_i(t)\right).$$

As $G(0) = 0$ and G is strictly increasing, $\sum_i q_i(t) \geq 0$. Hence the mean loss $\overline{L}(t) = -N^{-1}\sum_i q_i(t)$ is nonpositive; Definition 2.2 implies (3.4). ∎

The barriers hold for the entire admissible path without requiring terminal dollar closure. When the elasticity bound holds only near F, the argument applies while prices stay in that region; execution resources and network constraints may impose further restrictions. Section 4 compares how the attainable set changes across response functions.

3.1. Robustness to unmatched dollar flow and impact heterogeneity. The exclusion in Theorem 3.2 uses both a common price-response function and exact dollar pairing. The local logarithmic price response per executed dollar is $\lambda(S) = g(S)/S^2$, with elasticity

$$\frac{\mathrm{d}\log\lambda(S)}{\mathrm{d}\log S} = e(S) - 2.$$

Thus the threshold $e \geq 2$ states that this model's local per-dollar log-impact does not increase as the price falls. We quantify two departures from the hypotheses of the exclusion theorem while retaining fixed fundamentals and common price-level recovery.

Proposition 3.3 (Stability under unmatched dollar flow). *Suppose $S_i(0) = F$, all prices remain positive, and the homogeneous response $g > 0$ and common recovery rate $\rho > 0$ satisfy $e(S) \geq 2$ on an interval containing F and every attained price. For absolutely continuous executed-dollar flows, relax exact pairing to $\sum_i \dot{X}_i(t) = \varepsilon(t)$ with $\varepsilon \in L^1([0,T])$. Then, for all $t \in [0,T]$ and $\alpha \in (0,1]$,*

$$\begin{aligned}\mathcal{C}_\alpha(t) \leq \overline{L}(t) &\leq -\frac{g(F)}{NF^2}\int_0^t e^{-\rho(t-s)}\varepsilon(s)\,\mathrm{d}s \\ &\leq \frac{g(F)}{NF^2}\int_0^t e^{-\rho(t-s)}[-\varepsilon(s)]_+\,\mathrm{d}s.\end{aligned}$$

Proof. Let $H(t) = \sum_i \Phi(S_i(t))$, where $\Phi(F) = 0$ and $\Phi'(S) = S/g(S)$. Concavity of Φ, implied by $e \geq 2$, gives $(S-F)\Phi'(S) \leq \Phi(S)$. Hence

$$\dot{H} = \varepsilon - \rho\sum_i (S_i - F)\Phi'(S_i) \geq \varepsilon - \rho H,$$

and $H(0) = 0$ implies $H(t) \geq \int_0^t e^{-\rho(t-s)}\varepsilon(s)\,\mathrm{d}s$. Moreover, $G(q) = \Phi(Fe^q)$ is concave and $G'(0) = F^2/g(F)$, so its tangent inequality yields

$$H(t) = \sum_i G(q_i(t)) \leq \frac{F^2}{g(F)}\sum_i q_i(t) = -\frac{NF^2}{g(F)}\overline{L}(t).$$

Combining these bounds and discarding the nonnegative part of ε in the weighted integral proves the claim. ∎

Here ε measures the unmatched executed-dollar rate *within the selected asset set*; it need not represent a net cash flow into the wider market. Exact pairing recovers Theorem 3.2. The estimate does not require terminal dollar-ledger closure.

Proposition 3.4 (Weighted-flow bound for heterogeneous quadratic impact). *Suppose $S_i(0) = F$ and $g_i(S) = \eta_i(S/F)^2$, where $\eta_i > 0$, with common $F > 0$ and $\rho > 0$. For positive absolutely continuous trajectories set $\kappa_i = \eta_i/F^2$ and $\omega(t) = \sum_i \kappa_i \dot{X}_i(t)$. Then, for all $t \in [0,T]$ and $\alpha \in (0,1]$,*

$$\begin{aligned}\mathcal{C}_\alpha(t) \le \overline{L}(t) &\le -\frac{1}{N}\int_0^t e^{-\rho(t-s)}\omega(s)\,\mathrm{d}s \\ &\le \frac{1}{N}\int_0^t e^{-\rho(t-s)}[-\omega(s)]_+\,\mathrm{d}s.\end{aligned}$$

Writing $\bar{\kappa} = N^{-1}\sum_i \kappa_i$ and $\varepsilon = \sum_i \dot{X}_i$, the weighted imbalance decomposes as

$$\omega = \bar{\kappa}\,\varepsilon + \sum_i (\kappa_i - \bar{\kappa})\dot{X}_i.$$

Consequently exact dollar pairing alone need not eliminate the weighted flow when the impact strengths differ.

Proof. With $q_i = \log(S_i/F)$, the impact law becomes $\dot{q}_i = -\rho(1 - e^{-q_i}) + \kappa_i \dot{X}_i$. Jensen's inequality and $e^x \ge 1 + x$ imply

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\overline{L}(t) &= \rho\left(1 - \frac{1}{N}\sum_i e^{-q_i}\right) - \frac{\omega}{N} \\ &\le \rho(1 - e^{\overline{L}}) - \frac{\omega}{N} \le -\rho\overline{L} - \frac{\omega}{N}.\end{aligned}$$

The result follows by comparison from $\overline{L}(0) = 0$. ∎

A terminally dollar-closed counterexample. The loss of the homogeneous barrier is possible even with exact dollar pairing *and* terminal closure. Take $N = 2$, $\kappa_1 > \kappa_2 > 0$, and two matched executed-dollar pulses: $(\mathrm{d}X_1, \mathrm{d}X_2) = (a, -a)$ at time 0, followed by $(-a, a)$ at time $T > 0$. Both asset ledgers close exactly, and total paired-dollar turnover is $2a$. Writing $r = e^{-\rho T} \in (0,1)$, the terminal log prices are

$$\begin{aligned}q_1(T) &= \log\bigl(1 + r(e^{\kappa_1 a} - 1)\bigr) - \kappa_1 a, \\ q_2(T) &= \log\bigl(1 + r(e^{-\kappa_2 a} - 1)\bigr) + \kappa_2 a.\end{aligned}$$

For sufficiently small $a > 0$, Taylor expansion gives

$$\overline{L}(T) = \frac{(1-r)(\kappa_1 - \kappa_2)}{2}\,a + O(a^2) > 0, \qquad Q_{1/2}(T) = (1-r)\kappa_1 a + O(a^2) > 0.$$

Therefore $\mathcal{C}_{1/2}(T) > 0$, although each response has elasticity $e_i = 2$. Prices remain strictly positive, and a sufficient initial share inventory covers the finite sales. This construction uses

instantaneous pulses and does not impose a gross-rate cap. The common-impact hypothesis of Theorem 3.2 fails in this example.

The two perturbation estimates isolate departures from matching and homogeneous impact strength. They keep F and ρ fixed; time-varying fundamentals, heterogeneous restoration, and nonseparable cross-impact require separate analysis.

4. Reachability Regimes under Power-Law Impact.

Main results of this section. The power-law family separates structural barriers, finite-inventory severity bounds and constructive reachability. The distinction between partial and full coverage also yields different horizon and resource limits.

To compare exclusion and reachability under matched assumptions, we vary only the response shape in the family

$$g_p(S) = \eta \left(\frac{S}{F}\right)^p, \qquad p \in \{2, 1, 0\}, \quad \eta > 0. \tag{4.1}$$

All three markets have identical N, T, F, ρ, η, identical initial prices and initial *impact*, the same paired-dollar trading network, and, when compared numerically, the same turnover and inventory caps. They differ only in the exponent p. In the log-dollar equation (2.5),

$$D_{ii}^{(p)}(q_i) = \frac{\eta}{F^2} e^{(p-2)q_i}, \qquad e(S) = p. \tag{4.2}$$

The same coordinate formula integrates explicitly for every exponent:

$$\Phi_p(S) = \begin{cases} \dfrac{F^p}{\eta(2-p)}\left(S^{2-p} - F^{2-p}\right), & p \neq 2, \\ \dfrac{F^2}{\eta}\log(S/F), & p = 2. \end{cases}$$

Consequently, paired trades conserve a quantity proportional to $\sum_i S_i^2$, $\sum_i S_i$, or $\sum_i \log S_i$ for $p = 0, 1, 2$, respectively. Recovery evolves these quantities between trades. Figure 2 shows the corresponding instantaneous response per executed dollar: below F, the $p = 0$ response amplifies dollar trades most strongly.

Theorem 4.1 (A/B/C reachability regimes). *Let $N \geq 2$ and $k = \lceil \alpha N \rceil < N$. Compare the three markets in (4.1) under the same model protocol.*

(A) ***Barrier market*** *($p = 2$): $\mathcal{C}_\alpha(t) \leq 0$ for every admissible path and every horizon.*

(B) ***Inventory-bounded market*** *($p = 1$): positive but sufficiently small terminal systemic drawdowns are reachable. If every asset starts with a fixed finite inventory cap H_0 and short selling is forbidden, the terminal crisis severity is bounded above independently of the trade sequence.*

(C) ***Market with unbounded partial-coverage losses*** *($p = 0$): for every fixed $N \geq 2$ and $k < N$, there exists a sufficiently long but finite horizon T and finite constants M_0, H_0 such that, for* every *finite $\delta > 0$, an admissible positive-price path with $M \leq M_0$, $H_i^- \leq H_0$, and $X_i(T) = 0$ attains $\mathcal{C}_\alpha(T) > \delta$.*

The constants in (C) may depend on N, T, F, ρ, η, but not on δ. They do not remain uniform as $N \to \infty$.

Proof. (A) follows from Theorem 3.2. (B) follows from the two-pulse construction in Proposition 4.3 and the inventory inequality (4.3) below. (C) is established constructively in Appendix C: $N-1$ targets are driven arbitrarily close to zero, while one asset serves as a cash-flow compensator. The control uses positive prices, fully paired marginal-dollar executions, a partition of recovery time, and terminal closure of every X_i. Its explicit turnover and inventory bounds are independent of the target final price $\varepsilon \downarrow 0$. ∎

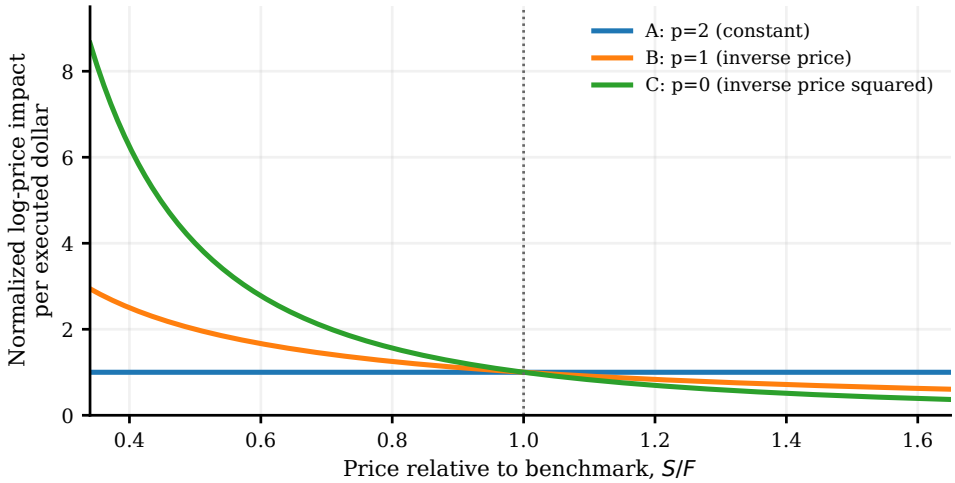


Figure 2. ***A single response family, three price sensitivities.*** *The plot shows the model-implied instantaneous* log-price change per executed dollar*, normalized by its common initial value. All curves coincide at $S = F$.*

The condition $k < N$ in (C) is essential: at least one asset must be available to carry the compensating flow. Finite gross share rates further constrain the compensating asset's *capacity.*

For $p = 1$, the inventory potential from (2.10) is

$$E_1(Fe^{-L}) = \frac{F^2}{\eta}\left(L + e^{-L} - 1\right). \tag{4.3}$$

By (2.11), terminal dollar closure and no short selling imply $E_1(S_i(T)) \le FH_0$. Since $L + e^{-L} - 1 \to \infty$ as $L \to \infty$, no fixed finite inventory can support arbitrarily severe losses of the requisite k assets. By contrast, for $p = 0$, $E_0(S) = (S-F)^2/(2\eta)$ remains bounded as $S \downarrow 0$; this makes the construction in (C) possible but does not by itself prove reachability.

Proposition 4.2 (Local all-asset decline). *If g is C^3 near F and $e(F) < 1$, then for every $N \ge 2$ there exists a terminally dollar-closed, positive-price, small-amplitude three-pulse control under which* every *$S_i(T) < F$. With $c = g(F)/F^2$ and $r = e^{-\rho T/2}$, the losses of a zero-sum impulse direction $\boldsymbol{u}$ with every $u_i \neq 0$ satisfy*

$$L_i(T) = \frac{1 - e(F)}{2} c^2 h^2 u_i^2 r(1 - r^2) + O(h^3) > 0 \tag{4.4}$$

for sufficiently small $h > 0$.

Proof. Apply the executed-dollar pulses $h\boldsymbol{u}$, $-(1+r)h\boldsymbol{u}$, $rh\boldsymbol{u}$ at $0, T/2, T$. They are pairwise dollar-balanced and sum to zero for every asset. Taylor expansion in the coordinate $y = \Phi(S)$ gives (4.4); details appear in Appendix B. ∎

Proposition 4.3 (Local partial-coverage decline). *Let $k < N$, $e_0 = e(F) < 2$, $c = g(F)/F^2$, and $r = e^{-\rho T} < 2 - e_0$. Select a zero-sum dollar direction with at least k positive components and execute $h\boldsymbol{u}$ at time 0 and $-h\boldsymbol{u}$ at time T. Then $\mathcal{C}_\alpha(T) > 0$ for all sufficiently small $h > 0$, and*

$$\overline{L}(T) = \frac{c^2 h^2}{2N}(1-r)(2-e_0-r)\sum_i u_i^2 + O(h^3). \tag{4.5}$$

Proof. The individual losses satisfy $L_i(T) = c(1-r)hu_i + O(h^2)$, while the first-order term vanishes when averaged because $\sum_i u_i = 0$. The strictly positive quadratic coefficient in (4.5) proves the result. See Appendix B. ∎

A concrete common-budget comparison. For $N = 3$, $k = 2$, $F = \eta = \rho = 1$, $T = 3$, common turnover cap $M_0 = 3$, and per-asset initial inventory cap $H_0 = 8.2$, the $p = 2$ market cannot reach any positive crisis; the $p = 1$ market can reach a small positive crisis but not the threshold $\delta = 10$ (it would require over 9 shares of selling by any qualifying asset); a constructive $p = 0$ path reaches $\mathcal{C}_{2/3}(T) \approx 13.32$ with turnover approximately 2.62 and a maximum per-asset cumulative share sale below 3.94. Here the large threshold $\delta = 10$ serves as a theoretical stress test separating bounded from unbounded logarithmic losses. In Market C, $E_0(Fe^{-\delta}) = \frac{F^2}{2\eta}(1-e^{-\delta})^2$ remains finite as $\delta \to \infty$, whereas $\log(F/S) \to \infty$ as $S \downarrow 0$. Thus, while all constructed paths maintain strictly positive prices, the unbounded-loss limit probes the asymptotic $S \downarrow 0$ boundary under normalized units ($F = \eta = 1$). Validation code accompanies the source files.

The common trading experiment across the three response regimes is illustrated in Figure 3.

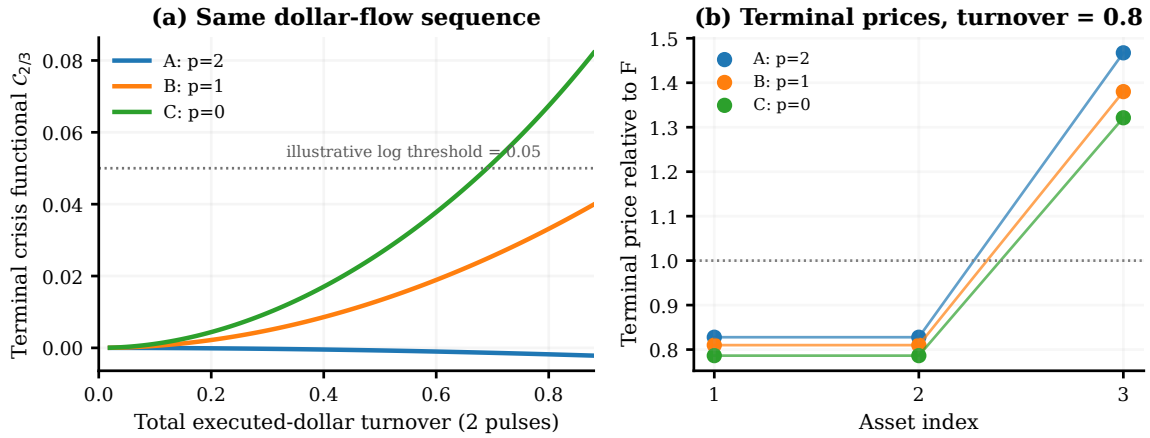


Figure 3. ***An identical trading experiment across A/B/C response functions.*** *Two terminally offsetting executed-dollar pulses act on $N = 3$ assets with $k = 2$, $F = \eta = \rho = 1$, and $T = 3$. Left: the measured* terminal *crisis functional as paired-dollar turnover increases; right: terminal prices under the same total turnover $M = 0.8$. The dotted horizontal line marks the reference log-loss threshold 0.05.*

The continuous power-law spectrum. The three canonical markets in Theorem 4.1 are part of a continuous family: retain the expression for g_p in (4.1), now allowing $0 \le p \le 2$. Since $e(S) = p$, the elasticity barriers and the local constructions distinguish two critical exponents. In geometric terms, $\Phi_p''(S) = (1-p)/g_p(S)$, while $G_p''(q) = F^2(2-p)e^{(2-p)q}/\eta$ for $G_p(q) = \Phi_p(Fe^q)$: the two curvatures reach their respective thresholds at $p = 1$ and $p = 2$. For $0 \le p < 2$, the dollar-coordinate distance to zero is finite, whereas the share-coordinate distance changes character at $p = 1$:

$$-\Phi_p(0^+) = \frac{F^2}{\eta(2-p)}, \qquad -\Psi_p(0^+) = \begin{cases} \dfrac{F}{\eta(1-p)}, & 0 \le p < 1, \\ +\infty, & 1 \le p < 2. \end{cases}$$

The inventory potential in (2.10), evaluated at $S = Fe^{-L}$, is for $p \neq 1, 2$

$$E_p(Fe^{-L}) = \frac{F^2}{\eta}\left[\frac{e^{(p-1)L} - 1}{p-1} - \frac{1 - e^{-(2-p)L}}{2-p}\right].$$

For $0 \le p < 1$, its zero-price limit is $E_p(0^+) = F^2/[\eta(1-p)(2-p)] < \infty$. Proposition 4.2 also makes a small, simultaneous decline of every asset locally reachable in this regime. Finiteness of the coordinate distances and inventory potential alone does not establish global reachability. Theorem 4.4 below supplies the required constructive extension of uniform-budget severe partial-coverage losses to every fixed $0 \le p < 1$.

At $p = 1$, Theorem 3.1 excludes simultaneous declines, while Proposition 4.3 gives local partial-coverage losses at every positive horizon. The potential (4.3) then grows linearly with L. For $1 < p < 2$, simultaneous declines remain excluded, but local partial-coverage losses are reachable under the sufficient horizon condition $e^{-\rho T} < 2 - p$, or $T > -\log(2-p)/\rho$. Here

the dollar distance is still finite while $E_p(Fe^{-L}) \sim F^2 e^{(p-1)L}/[\eta(p-1)]$ as $L \to \infty$. Indeed, if terminal executed-dollar flows vanish, short sales are prohibited, and the initial inventory of a qualifying asset is at most H_0, (5.2) implies

$$L \le \frac{1}{p-1} \log\left[1 + (p-1)\left(\frac{\eta H_0}{F} + \frac{1}{2-p}\right)\right], \qquad 1 < p < 2.$$

Thus, for fixed $p \in (1,2)$, the inventory-based severity ceiling scales at most logarithmically in large H_0, rather than linearly as for $p = 1$. At the second threshold $p = 2$, $-\Phi_2(0^+)$ diverges and Theorem 3.2 excludes every positive systemic crisis measure. These thresholds distinguish local and global reachability as well as the boundary resource singularities.

Construction of the severe-loss path. The generalized-radius construction closes each asset's executed-dollar ledger, not merely the market-wide sum, with finite turnover and inventory bounds. The full proof is in Appendix H.

Theorem 4.4 (Uniform-resource severe partial-coverage losses for sublinear impact). *Fix $N \ge 2$, $k = \lceil \alpha N \rceil < N$, positive F, η, ρ, and an exponent $0 \le p < 1$ in $g_p(S) = \eta(S/F)^p$. There exist a finite horizon $T > 0$ and finite constants M_0, H_0, depending on these fixed market parameters but not on the loss threshold, such that for every $\delta > 0$ an admissible positive-price, terminally dollar-closed path, using finitely many flow-completed pulses and absolutely continuous controls between them, satisfies*

$$\mathcal{C}_\alpha(T) > \delta, \qquad M(\gamma) \le M_0, \qquad H_i^-(\gamma) \le H_0 \quad (1 \le i \le N).$$

No constants uniform as $p \uparrow 1$ and no prescribed finite gross execution-rate cap are asserted.

Proof. Put $\nu = 2 - p > 1$. Paired-dollar pulses preserve $\sum_i S_i^\nu$, so their positive-price orbits are ℓ^ν spheres. The ℓ^ν-radial construction in Appendix H first closes the compensating asset's dollar ledger using a direction parameter $b_\varepsilon \in (b_-, \bar{b})$, where the positive lower bound $b_- > 0$ and the finite horizon are independent of ε. A partition by equal positive recovery-weight integrals then closes every target asset's ledger. On continuous control stages, all direction coordinates are at least $d_* = \min\{c, b_-\} > 0$, which bounds their share rates. On each of the finitely many transaction pulses, the gross share variation remains finite uniformly as $\varepsilon \downarrow 0$, because

$$\frac{F^p}{\eta} \int_0^{FN^{1/\nu}} u^{-p}\, \mathrm{d}u = \frac{F}{\eta(1-p)}\, N^{(1-p)/\nu} < \infty.$$

The ℓ^ν radius also bounds pulse-dollar variation; the appendix combines these estimates with bounded continuous turnover to give constants M_0, H_0 independent of ε and hence of δ. ∎

4.1. A sharp boundary when every asset must fall. Market C relies on $k < N$. To see why the breadth condition matters beyond a technical assumption, keep *the same constant-impact response* but change the target to $k = N$. There is then no nonqualifying asset available to compensate the other $N - 1$ terminal losses. In the sharp construction, the high-price role rotates among assets during recovery; finite recovery time limits how far their common terminal price can fall. The model therefore yields an exact horizon-dependent ceiling, even before a fixed execution budget is imposed.

Theorem 4.5 (Sharp all-asset horizon threshold). *Under Assumption 2.1 with $g(S) = \eta$ and $k = N$, define*

$$\delta_{\mathrm{crit}}(N,T) = \log \frac{\sqrt{N}}{1 + (\sqrt{N}-1)e^{-\rho T}}. \tag{4.6}$$

For every $\delta \in (0, \delta_{\mathrm{crit}}(N,T))$, some admissible terminally dollar-closed control reaches $\mathcal{C}_1(T) \geq \delta$ with finite turnover and sufficiently large finite inventory. If $\delta \geq \delta_{\mathrm{crit}}(N,T)$, no such positive-price path exists. Thus δ_{crit} is the unattained supremum *of the all-asset common log drawdown.*

Proof sketch. Set $R(t)^2 = \sum_i S_i(t)^2$. Under constant response, any paired-dollar trade satisfies $\mathrm{d}R^2|_{\mathrm{trade}} = 2\eta \sum_i \mathrm{d}X_i = 0$. Between trades,

$$\dot{R} = -\rho\left(R - F\frac{\sum_i S_i}{R}\right).$$

Strict positivity and $N \geq 2$ give $\sum_i S_i/R > 1$ at each finite time. Comparing the radial equation with $\dot{r} = -\rho(r - F)$, starting from $R(0) = F\sqrt{N}$, yields

$$R(T) > F\{1 + (\sqrt{N}-1)e^{-\rho T}\}.$$

If all terminal prices are at most $Fe^{-\delta}$, then $R(T) \leq F\sqrt{N}e^{-\delta}$; combining the two inequalities proves the strict necessary ceiling.

For sufficiency, use the N permutations of a positive unit direction $(a, b, \ldots, b)$, with $0 < b < 1/\sqrt{N}$. Each spherical reorientation is a paired-dollar pulse, and matched continuous flows keep a chosen direction fixed during recovery. The difference between high- and low-coordinate recovery integrands is the strictly positive weight W_b in (D.3). Partitioning the horizon into N equal integrals of W_b makes every asset's cumulative recovery integral identical. As the initial and terminal prices are also identical, the ledger identity (C.7) makes all $X_i(T)$ equal; paired-dollar balance makes their sum zero, so each is zero. Finally, varying b between its two limiting endpoints continuously spans all common terminal prices strictly between the benchmark and the unattained radial floor. Appendix D gives the positive-price construction in full. ∎

Twin ten-asset benchmark A1: all prices fall by half. Take $N = k = 10$, $F = \eta = 1$, $\rho = 0.4$, $T = 40$. Appendix D gives a constructive family of strictly positive directions $(a, b, \ldots, b)$ whose high-price role rotates among the ten assets. Solving the terminal condition $R_b(40)/\sqrt{10} = 1/2$ gives, numerically, $b \simeq 0.0668267688732$ and $a \simeq 0.979697732292$. Partitioning the recovery interval into ten equal integrals of the weight (D.3) gives $X_i(40) = 0$ for each asset and

$$S_1(40) = \cdots = S_{10}(40) = \frac{1}{2}, \qquad \mathcal{C}_1(40) = \log 2.$$

Thus the *ordinary* decline is 50% per asset; $\log 2 \simeq 0.69315$ is the log-loss severity. The analytic-primitive, double-precision reconstruction reports paired turnover $M \simeq 34.37368048$, maximum gross cumulative sale shares $\max_i H_i^- \simeq 16.58848226$, and maximum per-asset terminal executed-dollar residual below 10^{-12}. All intermediate prices remain strictly positive.

The theorem gives $\delta_{\mathrm{crit}}(10, 40) \simeq 1.15129230$ for the same market, equivalent to an *ordinary* common price-decline supremum of approximately 68.3772%. That limiting decline is *not attained*; 50% lies strictly inside the attainable range. Both the direction root and all equal-integral cut times are regenerated by `verification/twin_ten_asset_benchmarks.py`, with a checked fixture at `data/twin_ten_asset_benchmarks.json`.

Twin ten-asset benchmark A2: nine severe losses. Keep constant impact and $N = 10$, require only $k = 9$ qualifying assets, and set $F = \eta = \rho = 1$, $T = 4$. The Appendix C construction uses one compensator, a low direction coordinate $c = 10^{-3}$, and target terminal prices $\varepsilon = e^{-L}$. The $L = 3, 5, 10, 20$ numerical reconstructions in Table 5 (Appendix F) show how the resulting systemic log loss, paired turnover and gross share sales depend on target severity. These reconstructions illustrate the severity-independent finite resource bound established for the constructed path family by Theorem 4.1(C). A1 and A2 have distinct horizons and recovery rates.

5. Execution Resources and Turnover Bounds.

The idealized constructions in Section 4 demonstrate reachability with finite execution resources (for example, $M \simeq 34.4$ in A1). We now establish general lower bounds on the dollar turnover and share sales required for a specified drawdown.

Theorem 5.1 (Execution-resource lower bounds). *Suppose an admissible terminal path satisfies $\mathcal{C}_\alpha(T) \geq \delta > 0$. Write $k = \lceil \alpha N \rceil$ and $s = Fe^{-\delta}$. Alongside Φ, define the monotone share-response coordinate $\Psi(S) = \int_F^S \mathrm{d}u/g(u)$. Then the necessary dollar-turnover and total share-sale bounds are simply the respective coordinate distances to s:*

$$M \geq -k\Phi(s) = k \int_s^F \frac{u}{g(u)}\,\mathrm{d}u, \qquad \sum_i H_i^- \geq -k\Psi(s) = k \int_s^F \frac{\mathrm{d}u}{g(u)}. \tag{5.1}$$

Moreover, if $X_i(T) = 0$, no short sales are allowed, and the initial inventory of any qualifying asset does not exceed H_i, then

$$\int_s^F \frac{F - u}{g(u)}\,\mathrm{d}u \leq FH_i. \tag{5.2}$$

These necessary lower bounds need not be sharp.

Proof. For a terminally qualifying asset, the final price is at most $s < F$. Both transformed coordinates increase with price, and recovery acts toward their zero values at F. Consequently the cumulative negative executed-dollar and share variations must be at least $-\Phi(s)$ and $-\Psi(s)$, respectively. Summing over k distinct qualifying assets proves (5.1) because sales by qualifying assets form a subset of total sales, and each paired execution is counted once in total sales. Equation (5.2) follows from the nonnegative right-hand side of (2.11) with $X_i(T) = 0$ and $\theta_i(0) - \theta_i(T) \leq H_i$. Lemma A.1 in Appendix A justifies the variation argument for continuous finite-variation controls and flow-completed transaction pulses. ∎

Geometrically, the resource bounds measure distances in the two monotone coordinates. Trading direction, recovery time, and the buying capacity of receiving assets impose further requirements; we turn to these execution restrictions next.

Replicating identical asset depths at fixed α and δ gives $k = \Theta(N)$, and hence $M_*^{\mathrm{end}} = \Omega(N)$ for reachable targets. This experiment holds depth per asset fixed, so aggregate depth grows with N.

The low-price integrals discriminate different kinds of resource scarcity:

$$\int_0^F \frac{u}{g(u)}\,\mathrm{d}u, \qquad \int_0^F \frac{\mathrm{d}u}{g(u)}. \tag{5.3}$$

For a local power law $g(u) \asymp u^p$, the dollar integral converges only if $p < 2$, and the share integral only if $p < 1$. These integral criteria separate resource feasibility from the structural and dynamic conditions for reachability.

Dimensionless execution scales. Let $g_F = g(F) > 0$ and define the *local benchmark dollar-impact scale* $D_F = F^2/g_F$. At the benchmark, a signed executed-dollar amount D_F corresponds to a unit first-order log-price displacement under the model's marginal response rule, providing an intrinsic model scale. The dimensionless controls are

$$\widehat{M} = \frac{M}{D_F}, \qquad \widehat{H}_i = \frac{g_F H_i}{F}, \qquad \widehat{v} = \frac{g_F \bar{v}}{\rho F}, \qquad \widehat{T} = \rho T.$$

For the power family $g_p(S) = \eta(S/F)^p$, the necessary dollar-turnover bound in (5.1) becomes

$$\widehat{M} \geq k \begin{cases} \dfrac{1 - e^{-(2-p)\delta}}{2-p}, & p \neq 2, \\ \delta, & p = 2. \end{cases}$$

In the ten-asset, two-session constant-impact certificate, $F = g_F = 1$, $\rho = 0.4$, $M < 3$, $\max_i H_i^- < 2$, and $\bar{v}_{\mathrm{used}} < 0.285$ imply $\widehat{M} < 3$, $\max_i \widehat{H}_i^- < 2$, and $\widehat{v}_{\mathrm{used}} < 0.7125$. Relating these dimensionless quantities to observed market depth, accessible inventory, or average daily volume requires asset-specific empirical calibration of g_F and ρ.

Turnover and quadratic dissipation. The coordinate-distance bounds of Theorem 5.1 can be complemented by an aggregate estimate for constant per-share impact. Define

$$\mathcal{V}_2(t) = \frac{1}{2}\sum_{i=1}^{N}\big(S_i(t)^2 - F^2\big).$$

Proposition 5.2 (Turnover–dissipation bound under constant impact). *Suppose $g(S) \equiv \eta > 0$ and $S_i(0) = F$. Let the paired-dollar controls be absolutely continuous and assume that the entire price path satisfies $S_i(t) \geq a > 0$ and $|S_i(t) - F| \leq b$ for all i and $t \in [0, T]$. With $\dot{M}(t) = \sum_e \dot{z}_e(t)$ and $\mathrm{e} = \exp(1)$, the quadratic potential obeys the exact identity*

$$-\mathcal{V}_2(T) = \frac{\rho}{2}\int_0^T e^{-\rho(T-t)} \sum_{i=1}^{N}(S_i(t) - F)^2\,\mathrm{d}t \geq 0 \tag{5.4}$$

and the turnover inequality

$$-\mathcal{V}_2(T) \leq \frac{\rho\eta b}{a}\int_0^T (T-u)e^{-\rho(T-u)}\dot{M}(u)\,\mathrm{d}u \leq \frac{\eta b}{a\,\mathrm{e}}M(T). \tag{5.5}$$

Proof. Set $x_i = S_i - F$. The price equation $\dot{S}_i = -\rho x_i + \eta \dot{X}_i / S_i$ and dollar pairing $\sum_i \dot{X}_i = 0$ give

$$\dot{\mathcal{V}}_2 = -\rho \mathcal{V}_2 - \frac{\rho}{2} \sum_i x_i^2, \qquad \mathcal{V}_2(0) = 0.$$

Variation of constants proves (5.4). Applying it to the level-price equation and using $S_i \geq a$ yields

$$\begin{aligned} \sum_i |x_i(t)| &\leq \frac{\eta}{a} \int_0^t e^{-\rho(t-u)} \sum_i |\dot{X}_i(u)| \, \mathrm{d}u \\ &\leq \frac{2\eta}{a} \int_0^t e^{-\rho(t-u)} \dot{M}(u) \, \mathrm{d}u. \end{aligned}$$

The second inequality follows from $\sum_i |\dot{X}_i| \leq 2 \sum_e \dot{z}_e = 2\dot{M}$; oppositely directed flows at an asset can cancel in the net dollar rate. Since $\sum_i x_i^2 \leq b \sum_i |x_i|$, substitution into (5.4) and interchange of the nonnegative integrals give the first inequality in (5.5). The second follows from $\sup_{s \geq 0} \rho s e^{-\rho s} = \mathrm{e}^{-1}$. ∎

The time weight $\rho(T-u)e^{-\rho(T-u)}$ in (5.5) vanishes at $u = T$ and reaches its unique maximum at $T - u = 1/\rho$; it also decays for trades made far before the terminal date. This describes the contribution of a dollar of turnover to the *upper-bound estimate*: late trades allow little time for recovery dissipation, while earlier disturbances decay before T.

Two-session rolling-band example. Consider a uniform symmetric ±25% price band during each of two sessions, with the second session referenced to the preceding close. Then every price lies in $[9F/16, 25F/16]$, so $a = b = 9F/16$ is admissible and

$$-\mathcal{V}_2(T) \leq \frac{\eta}{\mathrm{e}} M(T).$$

If all N assets close at or below $(1-d)F$, $0 < d < 1$, then $-\mathcal{V}_2(T) \geq NF^2(2d - d^2)/2$, implying

$$M(T) \geq \frac{\mathrm{e}\, NF^2}{2\eta}(2d - d^2).$$

At $N = 10$, $F = \eta = 1$ and $d = 0.02$, this becomes $M(T) \geq 0.5382$, compared with $M(T) \geq 0.198$ from Theorem 5.1. Both bounds remain below $M = 3$.

For partial coverage, appreciation of the compensating assets can offset part of the falling subset's potential deficit. The aggregate potential itself cannot become positive by (5.4); without further constraints on compensating prices, it need not provide a useful positive lower bound on turnover for a specified falling subset alone.

6. Reachability under Bounded Execution Rates. A structurally reachable loss may require trades that cannot be completed at finite speed. We therefore restrict the control set by gross executed-share rates and single-session price bands. These restrictions preserve the barrier theorems, while positive reachability requires a new continuous construction rather than the pulse controls used in Market C. The B1–B2 witnesses in Section 7.9 belong to this restricted class, unlike A1–A2.

Finite-speed controls connect the execution setting to optimal trading and bounded-rate strategies [5, 33]. Here the rate limit constrains which aggregate loss states can be reached.

6.1. Two separate execution restrictions. For absolutely continuous nonnegative paired-dollar edge rates $\dot{z}_e(t)$, define an asset's *gross executed share rate* by

$$r_i^{\mathrm{gross}}(t) = \frac{1}{S_i(t)} \sum_e |B_{ie}| \dot{z}_e(t), \qquad 0 \le r_i^{\mathrm{gross}}(t) \le \bar{v}, \qquad i = 1, \dots, N. \tag{6.1}$$

This bound counts buy and sell executions *before* netting. In particular $|\dot{\theta}_i| \le r_i^{\mathrm{gross}} \le \bar{v}$ and instantaneous pulses are inadmissible. Bounding only the signed net share rate would allow simultaneous offsetting gross transactions of arbitrary size and would not constrain executed turnover.

Independently, an abstract *one-session price-band constraint* restricts every executed price to

$$F(1 - \beta_-) \le S_i(t) \le F(1 + \beta_+), \qquad 0 < \beta_- < 1, \quad \beta_+ > 0. \tag{6.2}$$

The control must keep the executed-price path within this interval; trajectories crossing the boundary are inadmissible. We treat (6.2) as an abstract state constraint and defer institutional comparisons with exchange price-limit rules to Section 8.

Proposition 6.1 (Rate and price-band exclusion certificates). *Suppose the common impact function is positive and $g(S) \le G$ on the whole admissible price region. Under the cap* (6.1)*, every asset obeys*

$$S_i(t) \ge F - \frac{G\bar{v}}{\rho}(1 - e^{-\rho t}), \qquad 0 \le t \le T. \tag{6.3}$$

Under the additional price band (6.2)*, put*

$$S_{\mathrm{floor}} = \max\left\{ F(1 - \beta_-),\ F - \frac{G\bar{v}}{\rho}(1 - e^{-\rho T}) \right\} > 0. \tag{6.4}$$

Then for any breadth α and any admissible control,

$$\mathcal{C}_\alpha(t) \le \log \frac{F}{S_{\mathrm{floor}}} \quad \text{for all } t \le T. \tag{6.5}$$

In particular, a necessary finite-speed condition to reach threshold $\delta > 0$ is

$$\bar{v} \ge \frac{\rho F(1 - e^{-\delta})}{G(1 - e^{-\rho T})}. \tag{6.6}$$

A one-session lower price limit independently excludes any $\delta > -\log(1 - \beta_-)$. These are necessary *constraints, generally not sharp reachability thresholds. For comparison, if a* hypothetical *multi-session regime resets a proportional lower band to each preceding close, then over D sessions the elementary bound is only $S_i(T_D) \ge F \prod_{d=1}^{D}(1 - \beta_{-,d})$ and $\mathcal{C}_\alpha(T_D) \le -\sum_{d=1}^{D} \log(1 - \beta_{-,d})$. The cumulative ceiling depends on the successive closing-price resets.*

Proof. The variation-of-constants formula for $\dot{S}_i = -\rho(S_i - F) + g(S_i)\dot{\theta}_i$ and $|\dot{\theta}_i| \le \bar{v}$ yields $S_i(t) \ge F - G\bar{v}\int_0^t e^{-\rho(t-s)}\,\mathrm{d}s$, proving (6.3). The price band supplies the second floor; every individual log loss is at most $\log(F/S_{\mathrm{floor}})$, and so are its cross-sectional mean, upper-breadth threshold, and their minimum. Rearranging the rate-only floor is necessary for an individual loss to reach δ. ∎

Dollar pairing and terminal closure yield additional market-wide execution requirements. Since each paired dollar is counted twice in gross share execution,

$$M = \frac{1}{2}\int_0^T \sum_i S_i(t) r_i^{\mathrm{gross}}(t)\,\mathrm{d}t \le \frac{NF(1+\beta_+)\bar{v}T}{2}. \tag{6.7}$$

Together with Theorem 5.1, this gives the necessary joint constraint

$$\bar{v} \ge \frac{2k}{NF(1+\beta_+)T}\int_{Fe^{-\delta}}^{F} \frac{u}{g(u)}\,\mathrm{d}u. \tag{6.8}$$

For a closed executed-dollar ledger, the nonnegative cost identity (2.11) also yields

$$E(S_i(T)) \le F\bar{v}T, \qquad \bar{v} \ge \frac{E(Fe^{-\delta})}{FT} \quad \text{for every qualifying asset.} \tag{6.9}$$

Bounds (6.6), (6.8), and (6.9) impose complementary restrictions on execution rates.

A per-asset rate estimate gives a first-order $O(\bar{v})$ bound on systemic losses. Under dollar pairing, the leading-order price movements cancel in the market mean, yielding the quadratic small-rate bound below for constant impact.

Theorem 6.2 (A quadratic small-rate crisis barrier for constant impact). *Let $g(S) \equiv \eta > 0$, assume dollar pairing $\sum_i S_i\dot{\theta}_i = 0$, and impose the gross share-rate cap (6.1). Write $d_T = 1 - e^{-\rho T}$ and*

$$b_T = \frac{\eta\bar{v}}{F\rho}\,d_T < 1. \tag{6.10}$$

Then, with or without terminal dollar closure and for every $N \ge 2$ and breadth α,

$$\mathcal{C}_\alpha(T) \le \left(\frac{\eta\bar{v}}{F\rho}\right)^2 \left[d_T - \rho T e^{-\rho T} + \frac{d_T^2}{2(1-b_T)^2}\right]. \tag{6.11}$$

Consequently, the supremum of attainable systemic losses is $O(\bar{v}^2)$ as $\bar{v} \downarrow 0$. For fixed N, T and positive price-band widths on both sides of the benchmark, Theorem 6.3 supplies a matching construction. Here $\bar{v}_*(\delta)$ denotes the infimum rate cap permitting severity at least δ, with fixed $N, T, F, \rho, \eta, \alpha$, positive per-asset initial inventory allowances, terminal dollar closure, and prescribed bands $\beta_- > 0$, $\beta_+ > 0$. In this constrained class, $\bar{v}_*(\delta) = \Theta(\sqrt{\delta})$ as $\delta \downarrow 0$. These bounds determine the scaling order; the optimal constants remain undetermined.

Proof. Set $x_i(t) = S_i(t) - F$ and $m(t) = \sum_i x_i(t)$. Variation of constants gives $|x_i(t)| \le \eta\bar{v}(1 - e^{-\rho t})/\rho$. Since $\sum_i S_i\dot{\theta}_i = 0$, $\sum_i \dot{\theta}_i = -F^{-1}\sum_i x_i\dot{\theta}_i$ and therefore $\dot{m} = -\rho m - (\eta/F)\sum_i x_i\dot{\theta}_i$. Integrating and bounding the gross rates yields

$$\frac{|m(T)|}{NF} \le \left(\frac{\eta\bar{v}}{F\rho}\right)^2 [d_T - \rho T e^{-\rho T}]. \tag{6.12}$$

For $|x_i(T)|/F \le b_T < 1$, Taylor's inequality $-\log(1+y) \le -y + y^2/[2(1-b_T)^2]$ implies $\overline{L}(T) \le |m(T)|/(NF) + b_T^2/[2(1-b_T)^2]$. Use $\mathcal{C}_\alpha \le \overline{L}$ to obtain (6.11). The construction in Theorem 6.3 has $\mathcal{C}_1(T) = c_N a^2 + O(a^3)$ with $c_N > 0$, and its gross rate is $O(a)$; setting $a \propto \bar{v}$ gives a positive loss of matching order $\Omega(\bar{v}^2)$ while keeping a fixed positive band. This proves the stated small-rate exponent. ∎

Theorem 6.3 (Positive all-asset decline with bounded continuous execution). *Assume $g \in C^3$ near F, $g(F) > 0$, $\rho > 0$, and $e_0 = Fg'(F)/g(F) < 1$. Fix* any *$N \ge 2$, horizon $T > 0$, rate cap $\bar{v} > 0$, one-session band widths $\beta_-, \beta_+ > 0$, and any fixed positive amount of initially available shares per asset. Then there exists an* absolutely continuous *paired-dollar control with per-asset gross share rate at most $\bar{v}$, remaining strictly inside the band, with $X_i(T) = 0$ for each asset and*

$$S_i(T) < F \text{ for every } i, \qquad \mathcal{C}_1(T) > 0. \tag{6.13}$$

The attained loss can be arbitrarily small.

Proof. Choose a zero-sum vector $u \in \mathbb{R}^N$ with every $u_i \neq 0$. Put $\Delta = T/3$, $r = e^{-\rho\Delta}$ and set the bounded, three-bin signal

$$f(t) = \begin{cases} 1, & 0 \le t < \Delta, \\ -(1+r), & \Delta \le t < 2\Delta, \\ r, & 2\Delta \le t \le T, \end{cases} \qquad \dot{X}_i = a u_i f(t). \tag{6.14}$$

The dollar rates sum to zero, and $\int_0^T f(t)\,\mathrm{d}t = 0$.

Direct integration also gives $\int_0^T e^{-\rho(T-t)} f(t)\,\mathrm{d}t = 0$. Write $h(S) = g(S)/S$ and expand $S_i(t) = F + a u_i y(t) + a^2 u_i^2 z(t) + O(a^3)$, where $\dot{y} = -\rho y + h(F) f$, $y(0) = 0$, and $\dot{z} = -\rho z + h'(F) f y$, $z(0) = 0$. The weighted cancellation implies $y(T) = 0$. Since $h'(F)/h(F) = (e_0 - 1)/F$, integration by parts yields

$$\log \frac{F}{S_i(T)} = \frac{\rho(1-e_0)}{2F^2}\, a^2 u_i^2 \int_0^T e^{-\rho(T-t)} y(t)^2\,\mathrm{d}t + O(a^3). \tag{6.15}$$

The integral is strictly positive. Thus all prices fall for every sufficiently small $a > 0$. Represent each zero-sum dollar-rate vector by nonnegative source-to-destination edge rates with minimum gross flow, so $r_i^{\mathrm{gross}} = |a u_i f|/S_i$. The entire price path converges uniformly to F as $a \downarrow 0$, ensuring compliance with the rate cap, price band, and any positive inventory allowance for sufficiently small a. ∎

The continuous construction establishes local common-loss reachability under a finite rate cap and terminal dollar closure.

Interaction of rate and price-band constraints. The global barrier for $e(S) \ge 2$ and the full-coverage exclusion for $e(S) \ge 1$ remain valid after restricting the set of admissible controls. For $p = 1$, a bounded two-stage buy-then-sell dollar-rate control still makes $k < N$ assets end below F: its per-asset signed dollars integrate to zero, while $\sum_i S_i(T) = NF$ and strict concavity of log makes the average log loss positive. In contrast, the *arbitrarily severe* construction for $p = 0$ in Theorem 4.1(C) uses instantaneous sweeps and therefore falls outside the rate-constrained

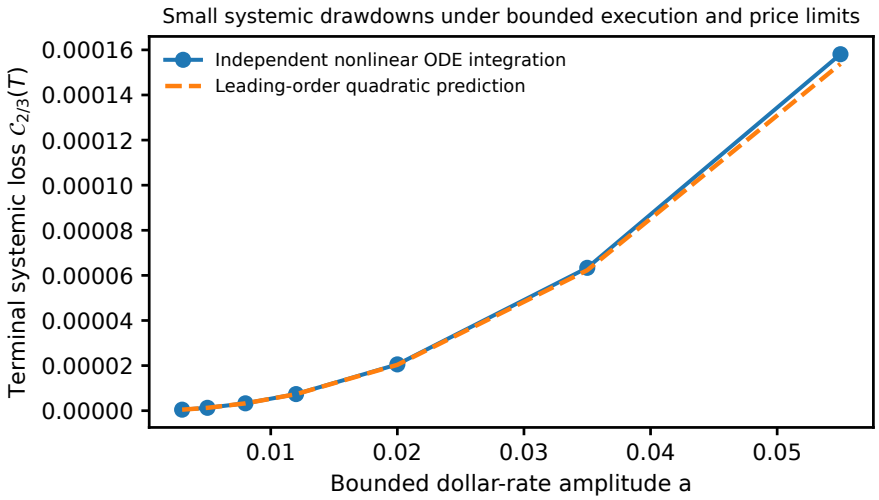


Figure 4. ***Quadratic local all-asset loss under a finite-rate control.*** *Independently integrated nonlinear price paths for constant g agree with the leading-order quadratic prediction from* (6.15) *as the amplitude decreases. The displayed functional is $\mathcal{C}_{2/3}(T)$, evaluated for the all-asset-decline construction of Theorem 6.3. The plotted amplitude is sufficiently small to preserve positive prices; the admissibility checks for the highlighted $a = 0.035$ case are reported in the code bundle.*

class (6.1). Under a fixed price band, no one-session terminal loss can exceed $-\log(1-\beta_-)$. Theorem 6.3 establishes strictly positive local all-asset declines under both restrictions, with severity limited by the rate and price-band constraints.

Table 1

How execution restrictions affect reachability for $p = 0$. Arbitrarily severe partial-coverage losses use a sufficiently long fixed finite horizon, as in Theorem 4.1(C). Local reachability holds for every $T > 0$; "local" refers to a sufficiently small positive terminal all-asset loss.

Admissible controls	Established reachability	Scope of the result
Instantaneous sweeps, no band	Arbitrarily severe partial-coverage log loss, $k < N$	Does not model a bounded execution rate
Gross rate $\bar{v}$ only	Local all-asset loss for any $\bar{v} > 0$	Unbounded severe-loss construction not certified
Hard session band only	Local all-asset loss when $\beta_- > 0$ and $\beta_+ > 0$	$\delta > -\log(1-\beta_-)$ impossible
Both restrictions	Local all-asset loss for positive caps/widths; rate and price-floor certificates	No general exact severe threshold or optimal execution rate

Figure 4 compares the numerical small-rate loss with its leading-order quadratic prediction.

7. Rolling Price Limits: Constrained Realizations. We apply a multi-session proportional-reset price-band protocol in which each session's admissible executed-price interval is referenced to the preceding close. The analysis links the obstruction to a second all-asset lower-limit close

with earlier above-benchmark price excursions and the capacity of the remaining assets to receive paired-dollar flow. The resulting two-session interval-certified constructions jointly test breadth, mean loss, gross rates, inventory, and rolling bands; further uniform-band comparisons and exact controls are retained in Appendix I. Trading interruptions can also redirect order pressure through time and across markets, as the circuit-breaker analysis of Subrahmanyam [31] illustrates.

Throughout, a *lower-band close* (also called a *lower-limit close* in some theorem statements) means that the terminal executed price attains the modeled session floor. Institutional features such as resting order queues or trading halts lie outside the executed-price state space (see Section 8).

7.1. Rolling bands and the exact cumulative accounting. Partition trading time into D sessions $[T_{d-1}, T_d]$, each of duration $\tau_d > 0$. Overnight calendar time does not contribute to the model clock; prices carry over continuously from the preceding close. In the stylized *proportional-reset* version, set $R_{i,d} = S_i(T_{d-1})$ and require

$$(1-\beta_{i,d,-})R_{i,d} \le S_i(t) \le (1+\beta_{i,d,+})R_{i,d}, \qquad t \in [T_{d-1}, T_d], \tag{7.1}$$

where $0 < \beta_{i,d,-} < 1$. A lower-limit close is $S_i(T_d) = (1-\beta_{i,d,-})R_{i,d}$; a D-day streak means this equality holds for the *same* assets in D consecutive sessions. A streak in $k = \lceil \alpha N \rceil$ assets is not yet a systemic crisis under Definition 2.2: the market-average log loss must also reach the chosen threshold.

For a common constant lower fraction $q = 1 - \beta_-$, each target asset has $S_i(T_D) = Fq^D$ and loss $L_i(T_D) = D\lambda$ with $\lambda = -\log q$. Conversely, even without a streak, the state constraint ensures

$$\mathcal{C}_\alpha(T_D) \le D\lambda \tag{7.2}$$

for a common band and initial equal benchmark price. For heterogeneous bands the right side can be replaced by $\max_i \sum_{d=1}^{D} [-\log(1-\beta_{i,d,-})]$. These geometric ceilings precede the execution constraints examined below.

7.2. Cross-sectional crisis threshold under repeated lower-band closes. Consecutive lower-limit closes by k stocks satisfy the breadth requirement, but do not automatically satisfy the mean-loss requirement: the remaining assets may appreciate as they receive matched dollars. This is the same $k < N$ compensation mechanism that permits Market C's partial-coverage outcomes, now seen from the opposite direction. The following exact endpoint identity shows when the falling majority becomes a crisis under the *unchanged* functional, without yet constructing the trades.

Proposition 7.1 (A conditional systemic-crisis trigger test). *Suppose the same $k = \lceil \alpha N \rceil < N$ assets close at Fq^D after D identical proportional-reset sessions, with $q \in (0,1)$. Let the remaining $N-k$ assets close at $Y_j > 0$, and write $\lambda = -\log q$. For any $0 < \delta \le D\lambda$, the* terminal *crisis threshold $\mathcal{C}_\alpha(T_D) \ge \delta$ holds if and only if*

$$\prod_{j=1}^{N-k} \frac{Y_j}{F} \le \exp\big(kD\lambda - N\delta\big). \tag{7.3}$$

If all other assets have individual log losses at most $D\lambda$ (or if $k > N/2$, so the $N-k$ other assets cannot fill the top k ranks), then $Q_\alpha(T_D) = D\lambda$ and the terminal crisis severity is exactly

$$\mathcal{C}_\alpha(T_D) = \min\left\{ D\lambda, \frac{kD\lambda - \sum_{j=1}^{N-k} \log(Y_j/F)}{N} \right\}. \tag{7.4}$$

In particular, with a single compensator ($k = N-1$), its terminal price must obey $Y/F \le \exp(kD\lambda - N\delta)$ if a crisis of threshold δ is to occur.

Proof. The k target assets have loss $D\lambda$, hence $Q_\alpha \ge D\lambda$. For $\delta \le D\lambda$, the coverage part of the crisis criterion is therefore automatic, and the remaining requirement is $\overline{L}(T_D) \ge \delta$. Substituting the terminal prices into $\overline{L}(T_D) = N^{-1}\sum_i \log[F/S_i(T_D)]$ gives (7.3). If every other loss is at most $D\lambda$, or if $N-k < k$, the kth-largest loss is exactly $D\lambda$, proving (7.4). ∎

A terminal-price illustration for the ten-asset certificate is included with the interval evidence in Appendix G.

7.3. A minimum speed for consecutive lower-band closes. A closing price at the lower limit must be reached in finite time despite recovery toward F. For $g(S) \equiv \eta$, start a session at price s and suppose its length is τ and its lower floor is $(1-\beta)s$. Even before asking who purchases the matched dollars, the gross share-rate cap bounds the fastest decline through $\dot z = -\rho(z-F) - \eta\bar v$, $z(0) = s$. This yields a necessary threshold but leaves the market-wide matching problem unresolved.

Proposition 7.2 (Rolling-limit shock-rate certificates). *In the constant per-share impact model, a lower-limit close in a session starting at $s > 0$ requires*

$$\bar v \ge \left[\frac{\rho}{\eta} \left(F - s + \frac{\beta s}{1 - e^{-\rho\tau}} \right) \right]_+ . \tag{7.5}$$

If the same asset closes at the same proportional lower limit on each of D sessions of common duration τ, with $s_d = Fq^{d-1}$, then

$$\bar v \ge \frac{\rho F}{\eta} \max_{1 \le d \le D} \left[1 + q^{d-1} \left(\frac{1-q}{1 - e^{-\rho\tau}} - 1 \right) \right]. \tag{7.6}$$

Moreover, when $\eta\bar v < \rho F$, all prices starting at F obey the perpetual rate-only *floor $S_i(t) \ge F - \eta\bar v/\rho$, implying the necessary integer-streak bound*

$$D \le \frac{\log(1 - \eta\bar v/(\rho F))}{\log q}. \tag{7.7}$$

The bound is understood only for $\eta\bar v < \rho F$; with a larger rate cap it supplies no finite-day obstruction.

Proof. Variation of constants and $\dot\theta_i \ge -\bar v$ imply $S_i(\tau) \ge F + (s-F)e^{-\rho\tau} - (\eta\bar v/\rho)(1 - e^{-\rho\tau})$. Requiring $S_i(\tau) \le (1-\beta)s$ yields (7.5); substitute $s = Fq^{d-1}$ for each day to obtain (7.6). The same scalar comparison from initial F gives $S_i(t) \ge F - (\eta\bar v/\rho)(1 - e^{-\rho t})$, hence the uniform floor; combine it with Fq^D and note $\log q < 0$ to deduce (7.7). ∎

7.4. Obstruction to universal consecutive lower-band closes. The all-asset case is especially restrictive: there is no complement capable of absorbing a net dollar flow. The same potential that gave the original exclusion barriers now rules out a second simultaneous decline under a simple below-benchmark band condition. This obstruction extends the breadth distinction in the A/B/C regimes to the entire market.

Theorem 7.3 (An obstruction to a second all-asset limit-down close). *Assume $g_i(S) > 0$, $\rho_i > 0$, unchanged fundamental benchmarks $F_i > 0$, and instantaneous interasset executed-dollar balance $\sum_i \mathrm{d}X_i = 0$. Suppose every asset closes the first session at $S_i(T_1) = q_i F_i < F_i$, and the* entire second-session admissible band *lies at or below its own benchmark,*

$$(1 + \beta_{i,2,+})q_i \le 1 \quad \textit{for every } i. \tag{7.8}$$

Then it is impossible for every *asset to close the second session strictly below its first close. In particular, simultaneous second-day lower-limit closes in all N assets are impossible. This holds regardless of the transaction-rate cap, terminal signed-dollar balances, or the functional form of the positive g_i.*

Proof. Use $\Phi_i'(S) = S/g_i(S) > 0$ and $H = \sum_i \Phi_i(S_i)$. Trade settlement preserves H, and recovery contributes $\dot{H} = -\sum_i \rho_i S_i(S_i - F_i)/g_i(S_i) \ge 0$ whenever every $S_i \le F_i$. Condition (7.8) keeps the full second-session price path in that region. Hence $H(T_2) \ge H(T_1)$. If every second close were below the preceding close, strict monotonicity of all Φ_i would instead give $H(T_2) < H(T_1)$, a contradiction. ∎

For the stylized symmetric proportional band, $q_i = 1 - \beta_i$ and $\beta_{i,2,+} = \beta_i > 0$, so $(1 + \beta_i)(1 - \beta_i) = 1 - \beta_i^2 < 1$ and the theorem applies. The theorem uses full execution, paired-dollar settlement, and fixed fundamentals. Extensions with unmatched orders, an external cash account, or changing fundamentals require separate analysis. It leaves open consecutive streaks in $k < N$ assets supported by offsetting dollar flows in the remaining assets.

The same model cannot represent a positive-duration *all-asset locked limit-down state* with every price fixed strictly below F_i: maintaining $\dot{S}_i = 0$ would require $\dot{X}_i = \rho_i S_i(S_i - F_i)/g_i(S_i) < 0$ for every asset, contradicting $\sum_i \dot{X}_i = 0$. A limit-hit closing print is instead permitted as an endpoint condition. Persistent unmatched sell queues require a new order-book state, not simply an extended price trajectory.

7.5. A positive multi-day resource and accounting constraint. Even when not all assets fall, each target that is below F faces upward recovery. Terminal dollar closure therefore requires a path-history resource, not just a final selling flow: earlier above-benchmark excursions must provide the positive recovery contribution that compensates later losses. This is another expression of the same potential introduced in Section 3, now integrated over successive trading sessions.

Proposition 7.4 (Above-benchmark excursion required for a dollar-closed limit-down streak). *Let an asset begin at F and finish D sessions at $s_D < F$, with $X_i(T_D) = 0$. Then its trajectory necessarily visits $S_i > F$ on a set of positive trading-time measure and satisfies*

$$\rho \int_0^{T_D} \frac{S_i(t)(S_i(t) - F)}{g(S_i(t))}\,\mathrm{d}t = \int_{s_D}^{F} \frac{u}{g(u)}\,\mathrm{d}u > 0. \tag{7.9}$$

If every session after the first keeps the asset at or below F, and the first-session upper band is $U_1 > F$, a necessary condition is

$$\rho\tau_1 \max_{F\le u\le U_1} \frac{u(u-F)}{g(u)} \ge \int_{s_D}^{F} \frac{u}{g(u)}\,\mathrm{d}u. \tag{7.10}$$

For constant impact $g = \eta$, and $s_D = Fq^D$, $U_1 = F(1+\beta_+)$, this simplifies to $\rho\tau_1 \ge (1-q^{2D})/[2\beta_+(1+\beta_+)]$. Writing $H_+ = \rho\tau_1\beta_+(1+\beta_+)$, whenever $2H_+ < 1$ this imposes the additional necessary streak-length bound

$$D \le \frac{\log(1-2H_+)}{2\log q}. \tag{7.11}$$

This bound is specific to terminal dollar closure and the absence of above-fundamental excursions in later sessions.

Proof. Integrate the identity $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i - \rho S_i(S_i - F)\,\mathrm{d}t/g(S_i)$ over the full horizon. Since $X_i(T_D) = 0$ and $\Phi(F) = 0$, the integral must equal $-\Phi(s_D) > 0$. Any portion of the trajectory at $S_i \le F$ contributes a nonpositive integrand, so the positive part must arise above F. If later sessions cannot rise above F, all positive contribution must be earned in session 1, whose supremum is bounded by the left side of (7.10). ∎

7.6. A necessary receiving-capacity bound for multiasset execution. For $k < N$, a subset's net executed-dollar outflow must be absorbed by the remaining assets. Partition the assets into a falling subset J and its complement J^c. On a complete trading graph with node-level gross-rate caps and no separate edge capacities, conservation and the receiving assets' prices and rates bound the dollar flow across this partition.

Proposition 7.5 (A market-cut capacity and consecutive-limit lower bound). *Fix a subset J of assets and write J^c for its complement. Under paired-dollar execution and the per-asset gross share-rate cap $\bar v$, every admissible absolutely continuous path satisfies, for almost every t,*

$$\left|\sum_{i\in J} \dot X_i(t)\right| \le \bar v \min\left\{\sum_{i\in J} S_i(t), \sum_{j\in J^c} S_j(t)\right\}. \tag{7.12}$$

Suppose that throughout a session $[T_{d-1}, T_d]$ of length τ_d, each $i \in J$ stays at or below its unchanged benchmark F_i and ends at $s_{i,d} < s_{i,d-1}$. Let $\Phi_i'(S) = S/g_i(S) > 0$ and put

$$K_{J,d} := \sum_{i\in J} \left[\Phi_i(s_{i,d-1}) - \Phi_i(s_{i,d})\right] > 0. \tag{7.13}$$

If all compensating assets satisfy $S_j(t) \le U_{j,d}$ during that session, then the session's paired executed-dollar turnover M_d and the required gross-share-rate cap satisfy

$$M_d \ge K_{J,d}, \qquad \bar v \ge \frac{K_{J,d}}{\tau_d \sum_{j\in J^c} U_{j,d}}. \tag{7.14}$$

When $J^c = \varnothing$, the hypothesized simultaneous strict decrease of all assets cannot occur on a session lying entirely below benchmarks.

Proof. Dollar pairing gives $\sum_{i\in J}\dot{X}_i = -\sum_{j\in J^c}\dot{X}_j$. In either set, $|\dot{X}_i| \le S_i r_i^{\mathrm{gross}} \le S_i\bar{v}$, proving (7.12). The price-response coordinate obeys $\Delta X_i = \Phi_i(s_{i,d}) - \Phi_i(s_{i,d-1}) + \rho_i \int_{T_{d-1}}^{T_d} S_i(S_i - F_i)/g_i(S_i)\,\mathrm{d}t$. The integral is nonpositive for $i \in J$, hence $\sum_{i\in J}\Delta X_i \le -K_{J,d}$. The net executed dollars entering J^c are therefore at least $K_{J,d}$. The total matched-dollar turnover must cover this flow, and the complement's gross-share capacity is no more than $\bar{v}\tau_d \sum_{j\in J^c} U_{j,d}$. These yield (7.14). If J^c is empty, there is no opposing flow. ∎

A nonempty complement can impose an additional rate requirement through its receiving capacity. Its appreciation also offsets the market-average log loss in Proposition 7.1, so feasibility depends on both dollar absorption and the resulting average loss.

For constant per-share impact $g_i \equiv \eta$, $k = |J|$ identical lower-limit assets, initial benchmark F, common lower-limit ratio $q \in (0,1)$, and $m = N - k$ compensators with common upper-band fraction $\beta_{c,+}$, the dth-day specialization is

$$\bar{v} \ge \frac{k}{m}\,\frac{F\left(q^{2(d-1)} - q^{2d}\right)}{2\eta\tau_d(1+\beta_{c,+})^d} \quad (d \ge 2), \tag{7.15}$$

provided all k target prices remain below F and each compensator's preceding reference price has evolved from F under the specified upper bands. The k/m multiplier makes the funding bottleneck explicit. The condition gives a *necessary* lower bound on the rate. For the ten-asset, two-day example ($k = 9, m = 1, q = 0.97, \beta_{c,+} = 0.25, \tau_d = 20$), it gives only $\bar{v} \gtrsim 0.0080$; the stronger systemic quadratic bound below is more informative for that particular configuration.

7.7. A certified two-session construction and bounds on the critical rate. We construct and certify a path satisfying all the preceding constraints. Fix $N = 10$, with nine identical target assets and one compensator, $F = \eta = 1$, $\rho = 0.4$, two sessions of length 20, target rolling bands $\pm 3\%$, compensator bands $\pm 25\%$, and terminal closure $X_i(40) = 0$ for every asset. The targets share one prescribed continuous price path $x(t)$; the compensator follows the scalar nonlinear ODE determined by paired dollars. This symmetry reduces the calculation to two price states and establishes feasibility for the specified ten-asset model. A reproducible fixed-point interval calculation certifies the nonlinear compensator dynamics and all execution constraints for the piecewise-linear target path.

Proposition 7.6 (A computer-assisted two-day rate certificate). *In the ten-asset specification above ($k = 9$, $F = \eta = 1$, $\rho = 0.4$, $\tau_1 = \tau_2 = 20$, target bands $\pm 3\%$, compensator bands $\pm 25\%$), there exists an absolutely continuous, fully paired-dollar path with all per-asset terminal signed-dollar ledgers exactly zero such that the same nine assets close at 0.97 and 0.9409, respectively, while*

$$\mathcal{C}_{0.9}(40) > 0.020, \quad \bar{v}_{\mathrm{used}} < 0.285, \quad M < 3, \quad H_i^- < 2 \quad (1 \le i \le 10). \tag{7.16}$$

Every executed price remains inside its rolling session band. Here M counts paired executed dollars once and H_i^- denotes gross shares sold, so initial holdings of two shares of every asset are sufficient without shorting.

Proof sketch. Let $r = 23/10$. Prescribe the common target price $x(t)$ as the continuous piecewise-linear interpolation of

$$(t,x) = (0,1), (r,h), (20-r,h), (20,0.97), (20+r,0.995), (40-r,0.995), (40,0.9409). \tag{7.17}$$

The first plateau h is the unique root near 1.02 of the exact net-dollar-closure equation

$$\begin{aligned} &101600000000h^2 - 97138000000h - 6646157699 = 0, \\ &1.02020225507905 < h < 1.02020225507908. \end{aligned} \tag{7.18}$$

The compensator $y(t)$ is the unique positive solution of $\dot{y} = -0.4(y-1) - 9u(t)/y$, $y(0) = 1$, where $u(t) = x(t)[\dot{x}(t) + 0.4(x(t)-1)]$. All ten executed-dollar flows sum to zero at each time, and the exact root equation gives $X_i(40) = 0$ for every target, hence also for the compensator. A validated interval integration gives

$$\begin{gathered} 1.2219388 < y(20) < 1.2225356, \quad 1.4047874 < y(40) < 1.4055543, \\ \sup_t 9|u(t)|/y(t) < 0.284481, \\ \int_0^{40} 9[u(t)]_+/y(t)\,\mathrm{d}t < 1.880, \qquad 9\int_0^{40} |u(t)|\,\mathrm{d}t < 2.984. \end{gathered} \tag{7.19}$$

The targets have even smaller gross rates and gross selling shares; sessionwise interval enclosures verify both rolling bands. Exact rational enclosures for $e^{0.2}$ verify $\overline{L}(40) > 0.020$, while the nine target losses are $-\log(0.9409) > 0.020$, proving (7.16). ∎

For this ten-asset, two-session instance with $k = 9$, let $\bar{v}_*$ be the infimum gross rate cap among *all* admissible paths also satisfying $\mathcal{C}_{0.9}(40) \geq 0.020$. The quadratic system-wide bound (6.11) implies for every $\bar{v} \leq 0.044$ (with $F = \eta = 1$, $\rho = 0.4$, $T = 40$):

$$\mathcal{C}_{0.9}(40) \leq (\bar{v}/0.4)^2 \left[1 + \frac{1}{2(1-\bar{v}/0.4)^2}\right] \leq \frac{1563441}{79210000} < 0.020.$$

Since the right-hand side of (6.11) is continuous in $\bar{v}$ near 0.044 and is strictly below 0.020 at that value, the exclusion extends to some rates above 0.044. Together with the feasible construction in Proposition 7.6, this yields the rigorous, though nonoptimal, bracket for the critical gross-rate cap:

$$0.044 < \bar{v}_* < 0.285. \tag{7.20}$$

The certified trajectory supplies a feasible upper rate, while the quadratic barrier gives a strict lower bound for the same market instance. Determining the exact critical rate and whether it is attained remains a constrained multiday optimal-control problem.

7.8. A complementary uniform-band existence result. A second constructive case uses uniform ±25% rolling bands for all ten assets, in contrast to the preceding path's narrower target-asset bands. The two sessions, response and recovery parameters, gross share-rate cap, inventory limits and per-asset dollar closure are unchanged. The full comparison matrix, including higher-turnover controls, appears in Appendix I.

Proposition 7.7 (An eight-asset large drawdown under uniform rolling bands). *Under the uniform ±25% two-session protocol above, there exists a positive-price, absolutely continuous paired-dollar path satisfying $X_i(40) = 0$ and $H_i^- < 2$ for every asset, together with*

$$S_i(40) < 0.892F \quad (1 \leq i \leq 8), \qquad \mathcal{C}_{0.8}(40) > -\log(0.97), \qquad M < 3, \quad \bar{v}_{\text{used}} < 0.285.$$

Proof. Under the exact rational schedule `R80-10`, the common target price $x(t)$ and compensator price $y(t)$ satisfy $\dot{x} = -0.4(x-1) + u(t)/x$ and $\dot{y} = -0.4(y-1) - 4u(t)/y$. Outward-rounded fixed-point interval arithmetic (Appendix I) encloses both trajectories on $[0, 40]$ and verifies all rolling-band, rate, turnover, and share-sale constraints. The certified endpoints satisfy $x(40) < 0.892$ and $x(40)^8 y(40)^2 < 0.97^{10}$. Finally, $\int_0^{40} u(t)\,\mathrm{d}t = 0$ ensures $X_i(40) = 0$ for every asset. ∎

The additional strictly enclosed price paths establish further breadth–severity combinations. Their schedules, resource budgets and conservative loss thresholds are reported in Table 7 and the supplementary table 8.

7.9. Further certified ten-asset drawdowns under rolling price limits. The following ten-asset controls share one synthetic execution protocol: $F = \eta = 1$, $\rho = 0.4$, $T = 40$ in two 20-unit sessions, uniform ±25% rolling bands for *each* asset, strictly positive prices, gross share rate below 0.285 per asset, cumulative gross sales below two shares per asset, and terminal closure $X_i(40) = 0$ for every asset. Importantly, their turnover budgets *differ*. Table 2 promotes the most informative comparisons from Appendix I; Proposition 7.6, which uses different target and compensator rolling bands, remains a separate certified result.

Table 2

Twin ten-asset benchmark B: constrained, terminally dollar-closed controls. *Target-asset and equal-weight geometric-index declines are conservative* strict ordinary percentage *thresholds. The turnover caps are also strict; M counts paired executed dollars once. Each fixed rational control is individually interval checked for endpoint losses, rates, inventory and rolling bands.*

Control	Coverage	Target fall	Geometric-index fall	Turnover cap
P60-20	6/10	$> 20.90\%$	$> 5.0\%$	$M < 3$
R70-20-M4	7/10	$> 20.00\%$	$> 6.0\%$	$M < 4$
R70-20-M5 (B1)	7/10	$> 20.20\%$	$> 6.0\%$	$M < 5$
F100-2-M7 (B2)	10/10	$> 2.00\%$	$> 2.0\%$	$M < 7$

Witness B1 establishes that ordinary price declines above 20.2% in seven assets and 6% in the geometric index remain reachable under bounded execution rates, uniform rolling bands, assetwise dollar closure, and paired turnover $M < 5$. The P60-20 and R70-20-M4 controls provide further breadth–severity combinations with tighter turnover caps of $M < 3$ and $M < 4$, respectively. Witness B2 complements these results with a certified all-asset decline above 2% under a larger $M < 7$ cap. The prescribed rational controls, strict interval bounds, and input hashes are recorded in `verification/loss_breadth/`.

8. Discussion.

8.1. Economic accounting and execution burden. Recovery has two roles: for the price-level dynamics in Assumption 2.1, it enters the curvature-based exclusion argument, but it also makes nontrivial terminal displacement possible under dollar closure. If $\rho = 0$, $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i$ and $X_i(T) = 0$ implies $S_i(T) = F$ (Appendix A). For $\rho > 0$, the identity (2.11) shows that the nonnegative terminal potential and cumulative recovery dissipation are financed by a reduction of net share inventory, valued at the fixed benchmark. For any terminally dollar-closed path γ,

define this benchmark-valued inventory consumption by

$$\mathcal{K}(\gamma) := F\sum_i \big[\theta_i(0)-\theta_i(T)\big] = \sum_i E(S_i(T)) + \rho\sum_i \int_0^T \frac{(S_i(t)-F)^2}{g(S_i(t))}\,\mathrm{d}t \geq 0.$$

The functional $\mathcal{K}(\gamma)$ records the benchmark value of inventory consumed by terminal displacement and recovery dissipation. It uses dollar closure and accounts for net share-inventory consumption; transaction fees, spreads, outside funding, and counterparty balance sheets enter through additional costs. The inventory bound in Section 5 motivates an associated resource-minimization problem.

As a direct bound for the ten-asset certificate, $E(S) = (S-1)^2/2$ and the certified terminal compensator lower endpoint $y(40) > 1.4047874$ give

$$\mathcal{K}(\gamma) \;\geq\; \sum_i E(S_i(40)) > \frac{9}{2}(1-0.9409)^2 + \frac{1}{2}(1.4047874-1)^2 > 0.0976.$$

The certified terminal interval supplies this bound; enclosing the recovery integral would sharpen it. Let $\Gamma_{\alpha,\delta}$ denote the admissible paths satisfying the terminal crisis, pairing, closure, and execution restrictions. A natural extension is to optimize

$$\mathcal{K}_{\min}(\alpha,\delta) = \inf_{\gamma\in\Gamma_{\alpha,\delta}} \mathcal{K}(\gamma).$$

Jointly minimizing this inventory cost, executed-dollar turnover, and the gross rate gives a constrained optimal-control problem.

8.2. Dependence on resilience, impact, and the loss metric. The elasticity barriers depend on *price-level* recovery, $\dot S = -\rho(S-F)$. To distinguish this model from recovery of order-book volume, define $V(S) = \int_F^S g(u)^{-1}\,\mathrm{d}u$ and alternatively prescribe $\dot V = -\rho V$. Then

$$\dot S\big|_{\mathrm{vol}} = -\rho g(S)V(S), \qquad \dot\Phi\big|_{\mathrm{vol}} = -\rho S V(S).$$

Viewed as a function of V, the dollar coordinate satisfies $\mathrm{d}\Phi/\,\mathrm{d}V = S$ and $\mathrm{d}^2\Phi/\,\mathrm{d}V^2 = g(S) > 0$. Strict convexity with $\Phi(F) = V(F) = 0$ implies $SV(S) > \Phi(S)$ for $S \neq F$. Thus order-book volume and transaction dollars have related but nonidentical recovery geometries for nonconstant g.

Several *necessary resource* bounds survive. Since volume-coordinate recovery still restores positive prices monotonically toward F and both Φ and V increase with S, the dollar-coordinate turnover distance and share-coordinate gross-sale distance of Theorem 5.1 remain necessary. The inventory inequality also follows by applying the same potential E from (2.10):

$$\int_0^T S_i\,\mathrm{d}\theta_i = F[\theta_i(T)-\theta_i(0)] + E(S_i(T)) + \rho\int_0^T (S_i-F)V(S_i)\,\mathrm{d}t, \qquad (S_i-F)V(S_i)\geq 0.$$

The corresponding dissipation is nonnegative because V has the sign of $S-F$. For constant $g\equiv\eta$, $V = (S-F)/\eta$ makes the two recovery dynamics identical, so every constant-impact result and its certified controls are unchanged. For nonconstant g, the structural Section 3

elasticity barriers may change substantially, and general-impact statements in Sections 6–7 must be reassessed under the modified drift. No new elasticity threshold is claimed here.

The local response elasticity $e(S) = Sg'(S)/g(S)$ is distinct from cumulative metaorder impact, which also depends on order size, execution history and replenishment [1, 10, 17].

The severity of Market C is also a boundary result: logarithmic loss diverges as a positive price tends to zero, while constant g yields finite coordinate distance and finite inventory potential at that boundary. Fixed strictly positive price floors, realistic depth deterioration, rate caps, and trade halts can all limit its quantitative significance without invalidating the mathematical existence theorem. Accordingly, the attainable-loss bounds in Section 6 describe the quadratic small-rate regime; reaching a fixed large threshold requires adequate execution capacity.

The target statistic is the geometric-index and coverage measure defined in Section 2. In particular, the $p = 1$ market can preserve the arithmetic price mean at F while a partial-coverage geometric-index loss is positive. Capitalization weighting, arithmetic aggregation, rolling peaks and other breadth rules define different target sets, not equivalent versions of the present crisis criterion.

8.3. Deterministic reachability and institutional scope. The barrier theorems exclude terminal price states for every admissible control of the specified deterministic market, whereas the constructions establish feasibility of particular terminal losses. Because the framework isolates mechanical reachability under fixed fundamentals, identifying equilibrium trading incentives or information-free episodes in transaction data requires separate game-theoretic and empirical models.

The certified two-session path illustrates a constrained realization: paired executed-dollar flows, inventory, gross rates and rolling bands hold along its full price trajectory. Appendix G records the certificate, graphical reconstructions and sensitivity comparisons. The Tokyo Stock Exchange instead uses tiered absolute-yen daily limits with conditional widening rules [22]; institutional studies of price limits provide additional context [13, 23].

8.4. Further directions. Nonlinear recovery operators raise the question of which restoring fields preserve the Φ-based invariant regions. General heterogeneous (g_i, ρ_i, F_i), changing network topology, and nonseparable cross-impact alter the scalar potential argument; Subsection 3.1 establishes one diagonal extension with common F and ρ. For invertible matrix impact $\mathrm{d}\boldsymbol{S} = \mathsf{G}(\boldsymbol{S})\,\mathrm{d}\boldsymbol{\theta}$, an additive transaction potential would depend on exactness of the one-form

$$\omega = \mathbf{1}^{\top} \operatorname{diag}(\boldsymbol{S})\mathsf{G}(\boldsymbol{S})^{-1}\,\mathrm{d}\boldsymbol{S}$$

and on the compatible sign of its restoring drift. Matrix symmetry alone is not sufficient. Cross-impact no-dynamic-arbitrage restrictions [2, 30] and empirical multivariate impact [7] provide separate benchmarks.

Joint optimization of inventory cost, turnover, and the minimum feasible gross rate would connect reachability with trading incentives, including strategic predation [9]. Proposition 5.2 aggregates across assets and does not use per-asset terminal dollar closure. When $X_i(T) = 0$,

the constant-impact dynamics also give the subset-level identity

$$\frac{F^2 - S_i(T)^2}{2} = \rho \int_0^T S_i(t)(S_i(t) - F)\,\mathrm{d}t \qquad \text{for every } i.$$

This identity isolates each falling asset and may yield tighter subset-level turnover bounds. The squared-price dynamics, $\mathrm{d}(S_i^2) = -2\rho S_i(S_i - F)\,\mathrm{d}t + 2\eta\,\mathrm{d}X_i$, also have a trade term without the denominator $1/S_i$; their restoring drift remains nonlinear. An upper price bound $S_i \le U_{\max}$ together with the gross share-rate cap gives $\dot{M} \le NU_{\max}\bar{v}/2$. For fixed positive turnover, this prevents concentration at the single peak of the time kernel in (5.5) and permits a tighter weighted-turnover estimate. These constraints suggest time-dependent resource bounds sensitive to the composition of selling and receiving assets. Empirical identification of local impact g, recovery, trade direction, and accessible inventories would require time-resolved transaction and order-book observations; market-microstructure invariance [26] provides a scaling benchmark. A stochastic flow or equilibrium-selection model could then place probabilities alongside the present deterministic reachability results.

9. Conclusion. We have characterized systemic drawdown reachability in a deterministic market with paired-dollar execution, fixed fundamentals and price-level recovery. A response-adapted potential yields global elasticity barriers and quantitative bounds when dollar flows are unmatched or impact strengths differ. Power-law response functions generate distinct reachability regimes: severe partial-coverage losses admit fixed finite resource bounds at any fixed sublinear exponent, while full-coverage losses face an exact unattained horizon ceiling. Turnover and inventory inequalities quantify execution requirements, and continuous controls show that the quadratic small-rate loss upper bound has a matching attainable order under constant impact. Certified rolling-band paths demonstrate the consistency of these constraints in specified finite-dimensional markets. The separation between structural exclusion, constructive reachability and execution resources defines the scope of the results and the starting point for extensions to more general impact and recovery laws.

Code and numerical data availability. The research code, numerical data, certificates, and reproduction instructions are publicly available at https://github.com/Siyuan-chat/market-crash-reachability. The unchanged original computational baseline is associated with commit `ea831c4`; the frozen software archive is separately identified by DOI 10.5281/zenodo.23428183. Appendices G and I describe the two families of interval certificates and their verification scripts; Appendix H provides the constructive proof for sublinear impact. Reproduction commands and software requirements are in `REPRODUCE.md`; the checksum manifest is `release/SHA256SUMS`. The underlying certificate was previously regenerated from source, with independent floating-point ODE and exact-rational consistency checks. The idealized ten-asset reconstructions (A1/A2) and regression fixtures are implemented in `verification/twin_ten_asset_benchmarks.py` and `data/twin_ten_asset_benchmarks.json` through double-precision evaluation of the analytic primitives. The constrained witnesses (B1/B2) are supported by the outward-rounded interval-certificate archive in `verification/loss_breadth/`.

Declaration on AI-assisted tools. AI-assisted tools were used for language editing, manuscript organization, and assisting with verification and figure-generation scripts. The author

takes full responsibility for all mathematical proofs, numerical certificates, and scientific content.

Appendix A. Settlement, coordinate changes, and variation bounds. The continuous response rule is interpreted along finite-variation execution paths. During an idealized pulse, physical time is frozen ($\mathrm{d}t = 0$) and the states move along the response ODE as the executing share size varies. The pulse's executed dollar transfer is the *integral* $\int S_i \,\mathrm{d}\theta_i$, not the beginning-of-pulse price times the final share increment. By continuity of $g > 0$, its endpoints can be joined by an admissible response path whenever the intervening price stays positive.

For one asset the differential rule reads

$$\mathrm{d}\theta = \frac{\mathrm{d}S + \rho(S - F)\,\mathrm{d}t}{g(S)}. \tag{A.1}$$

Decompose $S = F + (S - F)$ and integrate:

$$\int_0^T S \,\mathrm{d}\theta = F[\theta(T) - \theta(0)] + \int_F^{S(T)} \frac{u - F}{g(u)}\,\mathrm{d}u + \rho \int_0^T \frac{(S - F)^2}{g(S)}\,\mathrm{d}t,$$

where the initial price is F. This proves (2.11). For $S < F$ the primitive $E(S)$ in (2.10) is also nonnegative because the integration orientation reverses the sign of the negative integrand. Hence if $\theta(T) = \theta(0)$, the total executed cost cannot be negative.

Let $y = \Phi(S)$ and $z = \Psi(S)$, with

$$\Phi'(S) = \frac{S}{g(S)}, \qquad \Psi'(S) = \frac{1}{g(S)}.$$

Then

$$\mathrm{d}y = \mathrm{d}X - \rho\frac{S(S - F)}{g(S)}\,\mathrm{d}t, \qquad \mathrm{d}z = \mathrm{d}\theta - \rho\frac{S - F}{g(S)}\,\mathrm{d}t. \tag{A.2}$$

Lemma A.1 (One-sided variation under restorative drift). *For any admissible finite-variation execution path starting at $S_i(0) = F$, including graph-completed transaction pulses, let $\mathrm{Var}^-(X_i; [0, T])$ and $\mathrm{Var}^-(\theta_i; [0, T])$ denote cumulative negative variation along the completed paths. Then*

$$\big[-\Phi(S_i(T))\big]_+ \le \mathrm{Var}^-(X_i; [0, T]), \qquad \big[-\Psi(S_i(T))\big]_+ \le \mathrm{Var}^-(\theta_i; [0, T]).$$

Proof. Use an absolutely continuous graph parameter σ for the finite-variation trajectory, with nondecreasing physical time $t(\sigma)$ and $t' = 0$ along each instantaneous pulse. The flow-completed chain rule for (A.2) gives, almost everywhere on the completed path,

$$\begin{aligned} y' &= X_i' - \rho\,\frac{S_i(S_i - F)}{g(S_i)}\,t', \\ z' &= \theta_i' - \rho\,\frac{S_i - F}{g(S_i)}\,t'. \end{aligned}$$

Because Φ and Ψ are increasing and both vanish at F, either coordinate is negative exactly when $S_i < F$. On that negative half-line its recovery drift is nonnegative; at the origin the drift vanishes. Applying the Lipschitz negative-part function to the absolutely continuous completed coordinates therefore yields, almost everywhere,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}[-y]_+ \le [-X_i']_+, \qquad \frac{\mathrm{d}}{\mathrm{d}\sigma}[-z]_+ \le [-\theta_i']_+.$$

The same inequalities apply on pulse arcs, where $t' = 0$ and trading follows the prescribed within-pulse execution flow. Integration, using $y(0) = z(0) = 0$, proves both bounds. ∎

For a qualifying asset $S_i(T) \le s < F$, monotonicity gives $-\Phi(S_i(T)) \ge -\Phi(s)$ and $-\Psi(S_i(T)) \ge -\Psi(s)$. Moreover, each directed paired-dollar transfer contributes negative dollar variation at only its selling endpoint; hence $\sum_i \mathrm{Var}^-(X_i; [0,T]) \le M$, even when simultaneous edge flows cancel at an asset. The lemma gives the dollar and share distances in (5.1). Finally, if $X_i(T) = 0$ and short sales are forbidden, equation (2.11) implies

$$E(S_i(T)) \le F[\theta_i(0) - \theta_i(T)] \le FH_i.$$

Since $E(S)$ increases as S falls below F, $S_i(T) \le s$ implies $E(s) \le FH_i$. This completes the proof of Theorem 5.1.

When $\rho = 0$, $\mathrm{d}\Phi(S_i) = \mathrm{d}X_i$. Terminal dollar closure therefore implies $\Phi(S_i(T)) = \Phi(F) = 0$ and, by strict monotonicity, $S_i(T) = F$. Within the present separable response class, recovery makes the nontrivial dollar-closed endpoint effect possible.

Appendix B. Local two-pulse and three-pulse expansions. We document the expansion used in Propositions 4.3 and 4.2. Normalize local variations by $x = S/F - 1$ and $c = g(F)/F^2$. At $S = F$, the executed-dollar coordinate $y = \Phi(S)$ satisfies $\Phi(F) = 0$, $\Phi'(F) = F/g(F)$, and

$$\frac{\mathrm{d}S}{\mathrm{d}y} = \frac{g(S)}{S}, \qquad \frac{\mathrm{d}^2S}{\mathrm{d}y^2} = \frac{g(S)[Sg'(S) - g(S)]}{S^3}. \tag{B.1}$$

Hence, writing $e_0 = e(F)$, the price ratio after a small dollar-coordinate displacement y is

$$\frac{S}{F} = 1 + cy + \frac{1}{2}(e_0 - 1)c^2y^2 + O(y^3), \tag{B.2}$$

with constants uniformly bounded for paths in a compact neighborhood of F.

For the two-pulse experiment, put $v = hu_i$ and $r = e^{-\rho T}$. The first pulse moves the dollar coordinate from zero to v; recovery multiplies the resulting *price-level* displacement $S - F$ by r. Using (B.2) and its local inverse, the dollar coordinate immediately before the final pulse is

$$\Phi(S_i(T^-)) = rv + \frac{e_0 - 1}{2}\, c\, r(1 - r)v^2 + O(v^3).$$

The final pulse subtracts v, so that $y_i(T) = \Phi(S_i(T^-)) - v$. Inverting once more and expanding $-\log(1 + w) = -w + w^2/2 + O(w^3)$ gives

$$L_i(T) = c(1 - r)hu_i + \frac{c^2h^2}{2}\,\lambda(r, e_0)u_i^2 + O(h^3), \tag{B.3}$$

$$\frac{1}{N}\sum_i L_i(T) = \frac{c^2h^2}{2N}(1 - r)(2 - e_0 - r)\sum_i u_i^2 + O(h^3),$$

where the explicit individual second-order coefficient is $\lambda(r, e_0) = (1-r)(2-e_0-r)$. The linear term vanishes in the mean because $\sum_i u_i = 0$. If at least k components are positive, those assets have positive first-order losses and the kth-largest loss is positive for small h; the mean is positive when $r < 2 - e_0$.

For the three-pulse experiment, the dollar-coordinate impulses at $0, T/2, T$ have amounts hu_i, $-(1+r)hu_i$, and rhu_i with $r = e^{-\rho T/2}$. Direct substitution and second-order expansion yield

$$L_i(T) = \frac{1-e_0}{2} c^2 h^2 u_i^2 r(1-r^2) + O(h^3), \tag{B.4}$$

with first-order terms canceling exactly. If $e_0 < 1$ and every $u_i \neq 0$, all terminal losses are positive for sufficiently small h. Since every pulse amount vector is zero-sum and the amounts cancel componentwise over the full horizon, the construction also satisfies the paired-dollar and terminal dollar-closure rules. The initial inventory requirement and turnover are finite for fixed h.

Appendix C. Uniform-resource severe drawdowns for partial coverage. We prove Theorem 4.1(C) constructively. Fix $N \geq 2$, set $m = N-1$, and choose the constant-response model $g(S) = \eta > 0$. We may assume the target crisis threshold δ is large, since a path exceeding a larger threshold automatically exceeds every smaller one. It is enough to produce m terminal prices $\varepsilon \downarrow 0$ with one remaining compensator asset, because every $k < N$ satisfies $k \leq m$.

C.1. A radial representation of paired-dollar trading. For $g \equiv \eta$,

$$\mathrm{d}(S_i^2) = -2\rho S_i(S_i - F)\,\mathrm{d}t + 2\eta\,\mathrm{d}X_i, \qquad \mathbf{1}^\top \mathrm{d}\boldsymbol{X} = 0. \tag{C.1}$$

Let $R = (\sum_i S_i^2)^{1/2}$. A paired-dollar trade keeps R fixed; a positive-price change at fixed R is implementable as a finite-variation path in the positive orthant of that sphere. In particular, interpolate the squared prices $Y_i = S_i^2$ linearly between two positive vectors of the same sum: every intermediate $Y_i > 0$ and $\sum_i \mathrm{d}Y_i = 0$, so $\mathrm{d}X_i = \mathrm{d}Y_i/(2\eta)$ is an admissible paired-dollar transfer. The corresponding share increment is $\mathrm{d}\theta_i = \mathrm{d}S_i/\eta$.

Fix $\beta = 1/5$. Choose $c > 0$ sufficiently small that, for every $b \in [0, \beta]$, the quantities

$$a = \sqrt{1 - b^2 - (m-1)c^2}, \qquad H = a + (m-1)c + b \tag{C.2}$$

satisfy $a > \max\{b, c\}$ and $bH < 1$. For example, $0 < c < 1/(10m)$ suffices. The constructed path uses $b \in (0, \beta)$; the endpoint $b = 0$ is used only by continuous extension in the root-bracketing argument below. During stage $j \in \{1, \ldots, m\}$, fix the spherical direction

$$\boldsymbol{u}_j = (c, \ldots, \underbrace{a}_{j}, \ldots, c, b), \qquad \|\boldsymbol{u}_j\|_2 = 1. \tag{C.3}$$

Set $\boldsymbol{S}(t) = R(t)\boldsymbol{u}_j$ throughout the stage. To make it solve the price equations, use

$$R(t) = F\left[H + (\sqrt{N} - H)e^{-\rho t}\right], \qquad \dot{\theta}_i = \frac{\rho F}{\eta}\left(u_{j,i}H - 1\right). \tag{C.4}$$

Indeed, differentiating $S_i = u_{j,i}R(t)$ in the price equation gives (C.4) and

$$\dot{X}_i = \frac{\rho F R(t)}{\eta}\, u_{j,i}(u_{j,i}H - 1), \quad \sum_i \dot{X}_i = \frac{\rho F R(t)}{\eta}\left(H\sum_i u_{j,i}^2 - \sum_i u_{j,i}\right) = 0, \tag{C.5}$$

because $\sum_i u_{j,i}^2 = 1$ and $\sum_i u_{j,i} = H$.

C.2. Closing the compensator's dollar flow. At $t = 0$ begin with $F\mathbf{1}$ and apply a positive-price pulse at fixed radius $F\sqrt{N}$ to the first direction $\boldsymbol{u}_1$. Switch between stages $\boldsymbol{u}_j$ using similar fixed-radius pulses. At time T use a final fixed-radius pulse to reach

$$S_1(T) = \cdots = S_m(T) = \varepsilon, \quad S_N(T) = \sqrt{R(T)^2 - m\varepsilon^2}. \tag{C.6}$$

The terminal price vector is positive for all sufficiently small ε.

For constant g, integrating (C.1) gives the exact executed-dollar ledger

$$X_i(T) = \frac{S_i(T)^2 - F^2}{2\eta} + \frac{\rho}{\eta}\int_0^T S_i(t)[S_i(t) - F]\,\mathrm{d}t. \tag{C.7}$$

For the compensator, its stage weight is always b. Define

$$\mathcal{F}_\varepsilon(b) = \frac{R_b(T)^2 - m\varepsilon^2 - F^2}{2\eta} + \frac{\rho}{\eta}\int_0^T bR_b(t)[bR_b(t) - F]\,\mathrm{d}t, \tag{C.8}$$

where R_b depends continuously on b through (C.4). With c fixed, choose a sufficiently large finite T such that $\mathcal{F}_0(\beta) < 0$. Such a choice exists because $R_\beta(t) \to FH_\beta$ and $\beta H_\beta < 1$: the integral in (C.8) then has a strictly negative linear term in T, while the terminal term remains bounded. Here H_β denotes H evaluated at $b = \beta$. Fix this horizon. Since $R_0(T) > F$, we have $\mathcal{F}_\varepsilon(0) > 0$ for all sufficiently small $\varepsilon > 0$, whereas

$$\mathcal{F}_\varepsilon(\beta) = \mathcal{F}_0(\beta) - \frac{m\varepsilon^2}{2\eta} < 0.$$

Continuity therefore gives $b(\varepsilon) \in (0, \beta)$ for which

$$X_N(T) = \mathcal{F}_\varepsilon(b(\varepsilon)) = 0.$$

Both c and T are fixed independently of ε.

C.3. Closing every target asset's dollar flow. It remains to make the other m terminal dollar balances equal. Whenever an asset is in the high state a rather than the low state c, the difference in its recovery integrand is

$$W(t) = (a - c)R(t)\big[(a + c)R(t) - F\big]. \tag{C.9}$$

This is strictly positive: $(a+c)H = 1 + b(a-b) + mac + bc > 1$, and $R(t) \geq FH$. Consequently $W(t) > 0$ for $t \in [0, T]$. Choose a partition $0 = t_0 < \cdots < t_m = T$ such that

$$\int_{t_{j-1}}^{t_j} W(t)\,\mathrm{d}t = \frac{1}{m}\int_0^T W(t)\,\mathrm{d}t, \qquad j = 1, \ldots, m. \tag{C.10}$$

The continuous, strictly increasing cumulative integral makes this partition possible. Let target asset j occupy the high state in stage j and the low state otherwise. All target assets have the same initial price F, terminal price ε, and recovery integral in (C.7), hence identical terminal balances $X_i(T)$. Every pulse and every stage respects $\sum_i \mathrm{d}X_i = 0$; since $X_N(T) = 0$, the common target balance must also vanish. Therefore

$$X_i(T) = 0 \qquad \text{for every } i = 1, \dots, N. \tag{C.11}$$

C.4. Uniform finite turnover and inventory. The path remains in the positive sphere of radius at most $F\sqrt{N}$, so $0 < S_i \le F\sqrt{N}$ throughout. There are $m+1 = N$ fixed-radius pulses, and each pulse can be implemented by monotone componentwise interpolation in squared-price coordinates. Consequently the cumulative paired-dollar pulse turnover has an explicit, though loose, upper bound depending only on N, F, η. For example, a safe bound is $N^3F^2/(4\eta)$. During recovery-controlled stages, $\sum_i |\dot{X}_i|/2 \le \rho N F^2/\eta$, giving

$$M(\gamma) \le M_0 := \frac{N^3F^2}{4\eta} + \frac{\rho N F^2 T}{\eta}. \tag{C.12}$$

Likewise, share sales over N pulses plus the continuous stages are bounded by

$$H_i^- \le H_0 := \frac{NF\sqrt{N} + \rho F T(\sqrt{N}+1)}{\eta}, \tag{C.13}$$

independently of ε. Starting with at least H_0 shares of each asset rules out short sales. The compensator's price stays bounded and positive; its log loss remains finite. The other m losses are $\log(F/\varepsilon) \to \infty$. Hence their kth order statistic and the average loss both diverge when $k \le m$. Thus $\mathcal{C}_\alpha(T) \to \infty$ with fixed finite M_0, H_0, proving Theorem 4.1(C).

Appendix D. Construction proving the sharp full-coverage bound. The necessary radial inequality was established in the proof of Theorem 4.5. For sufficiency, we construct an admissible family approaching the bound. Fix $0 < b < 1/\sqrt{N}$, and choose

$$a = \sqrt{1-(N-1)b^2} > b, \quad H = a + (N-1)b, \quad R_b(t) = F\left[H + (\sqrt{N} - H)e^{-\rho t}\right]. \tag{D.1}$$

Use the N positive unit-vector directions given by permutations of $(a, b, \dots, b)$. At time zero pulse from $F\mathbf{1}$ to $R_b(0)\boldsymbol{u}_1$, then follow $S_i(t) = u_{j,i}R_b(t)$ using the same paired controls as in (C.4). When changing directions, use finite fixed-radius positive squared-price pulses. At time T, make a final pulse to the equal-price point

$$S_i(T) = \frac{R_b(T)}{\sqrt{N}}, \qquad i = 1, \dots, N. \tag{D.2}$$

For the recovery integrals to be identical for every asset, define

$$W_b(t) = (a-b)R_b(t)\big[(a+b)R_b(t) - F\big]. \tag{D.3}$$

Here $(a+b)H = 1 + Nab > 1$, so $W_b(t) > 0$ for $t \in [0, T]$. Split the interval into N parts with equal integrals of W_b. In each part a different asset is the sole high-price coordinate. All N

assets now have the same initial price, the same terminal price, and the same recovery integral in (C.7). Their final dollar balances are therefore identical; since all trades are paired, their sum is zero and each balance is exactly zero.

As $b \downarrow 0$, the terminal equal price in (D.2) tends to

$$\frac{F}{\sqrt{N}}\left[1 + (\sqrt{N} - 1)e^{-\rho T}\right],$$

whereas $b \uparrow 1/\sqrt{N}$ yields $S_i(T) \to F$. Continuity in b implies that every strictly intermediate common terminal price is reachable by a positive-price, finite-turnover, fully dollar-closed path. The corresponding log-loss range is precisely $0 < \delta < \delta_{\mathrm{crit}}(N, T)$, completing the sharpness proof. The endpoint δ_{crit} itself would require the limiting direction with zero components and is excluded by strict price positivity.

Appendix E. Bounded-rate control: analytic details and audit protocol. For the three-bin control (6.14), the identities $\int f = 0$ and $\int e^{-\rho(T-t)} f(t)\,\mathrm{d}t = 0$ follow from equal bin widths and the geometric ratio $r = e^{-\rho T/3}$. Write $h_0 = g(F)/F$ and $h_1 = g(F)(e_0 - 1)/F^2$. The perturbation equations are $y' = -\rho y + h_0 f$, $z' = -\rho z + h_1 f y$. Since $y(0) = y(T) = 0$ and $(e^{-\rho(T-t)})' = \rho e^{-\rho(T-t)}$, one integration by parts gives the exact identity

$$\int_0^T e^{-\rho(T-t)} f(t) y(t)\,\mathrm{d}t = \frac{\rho}{2h_0} \int_0^T e^{-\rho(T-t)} y(t)^2\,\mathrm{d}t. \tag{E.1}$$

The resulting coefficient is $\rho/2$. Substituting $h_1/h_0 = (e_0 - 1)/F$ and $L_i = -\log(S_i/F)$ proves (6.15). At $p = 1$ the quadratic coefficient vanishes; this is consistent with the all-asset barrier but does not exclude a partial-coverage loss under a different bounded signal. The independent verifier integrates the nonlinear ODE directly with the dollar-rate signal, computes gross share execution before netting, and compares the finite-amplitude loss with (6.15).

A high-precision reconstruction of the severe pulse construction confirms its terminal-dollar accounting. Bounded-rate controls constitute a more restrictive admissible class, whose local construction and barrier are established separately.

Appendix F. Numerical experiments and reproducibility. This appendix documents deterministic price paths constructed under the exact marginal-price settlement protocol. For the severe-crash construction we use $g(S) = 1$, $F = \eta = \rho = 1$. The code selects a positive price direction and numerically solves for the compensator direction and time partitions; between switches the radius and ledger have analytic primitives. At every transaction time the paired executed-dollar sum is zero, and at maturity *each* signed-dollar ledger closes to numerical accuracy.

Table 3

Constructed paths with severe partial-coverage losses. The reported turnover is sufficient for these paths; the minimum turnover is unknown. All rows use target log loss $L = 5$ and parameters normalized to unity.

N	k	T	$\mathcal{C}_\alpha(T)$	Turnover M	Max. sale shares
3	2	3	3.3168	2.6179	3.9320
12	11	4	4.5781	38.6861	8.0804
20	19	5	4.7477	104.1750	10.1906

Table 4 reports the effect of increasing the target log loss for $N = 3$, $k = 2$, and $T = 3$.

Table 4

Effect of increasing target log loss for the fixed $N = 3$, $k = 2$, $T = 3$ construction.

Target log loss L	Systemic loss $\mathcal{C}_{2/3}(T)$	Turnover M	Maximum cumulative share sales
3	1.9845	2.6128	3.9365
5	3.3168	2.6179	3.9320
10	6.6501	2.6179	3.9319
20	13.3168	2.6179	3.9319

The second idealized ten-asset family, summarized in Table 5, fixes $N = 10$, $k = 9$, $F = \eta = \rho = 1$, $T = 4$ and $c = 10^{-3}$. The compensator direction and the equal recovery-weight cut times are solved numerically using the analytic radial primitives. The $\mathcal{C}_{0.9}$ entries are log-loss values rather than ordinary percentage declines. Every row is checked for strict price positivity, paired dollar flows and terminal ledger closure.

Table 5

Twin ten-asset A2: nine-target log-loss severity. *L and $\mathcal{C}_{0.9}$ are in log-loss units, M counts paired dollars once, and gross share sales count sales before netting. Entries are double-precision evaluations of the analytic primitives in Appendix C; the last column records the numerical dollar-ledger closure residual.*

L	$\mathcal{C}_{0.9}(4)$	M	$\max_i H_i^-$	$\max_i \lvert X_i(4)\rvert$
3	2.69569	27.16667	7.72723	$< 3 \cdot 10^{-14}$
5	4.49449	27.16461	7.72644	$< 3 \cdot 10^{-14}$
10	8.99447	27.16458	7.72743	$< 3 \cdot 10^{-14}$
20	17.99447	27.16458	7.72748	$< 3 \cdot 10^{-14}$

Run `python verification/twin_ten_asset_benchmarks.py --check` to reproduce both idealized families and check the regression fixture; the `--json` flag prints all underlying cut times, prices and residuals.

The plotted experiment uses a finite, high-resolution sampling grid only to display continuous recovery segments; the underlying segment values and cash-flow integrals are evaluated using analytic expressions, and terminal closure is checked separately. For the $N = 12$, $L = 5$ construction, the maximum terminal dollar-ledger residual under double precision is less than 10^{-13} in the distributed script. These residuals provide an implementation-level check of the analytic closure construction in Appendix C.

The numerical panels use model-generated inputs. Implementation details and reproduction commands are provided in the repository documentation.

Appendix G. A reproducible interval certificate for the two-day construction. The certificate uses outward-rounded fixed-point intervals and Taylor enclosures, following standard validated-numerics principles [32]. The implementation `verification/certify_streak_interval.py` uses integer and rational arithmetic at scale $Q = 10^{32}$, corresponding to resolution Q^{-1}; every multiplication and division is rounded outwards. The rational endpoint bracket for

h in (7.18) has strictly opposite polynomial signs and the polynomial is strictly increasing on that bracket, so it contains the unique relevant exact root.

On each linear target segment, write $x(t) = a + vt$ in local segment time and $u(t) = x(t)[v + \rho(x(t) - 1)]$. The compensator equation is the scalar nonautonomous ODE $\dot{y} = f(t, y) := -\rho(y - 1) - 9u(t)/y$. Its derivatives away from the finite set of target slope-change times are

$$f_y = -\rho + \frac{9u}{y^2}, \qquad f_t = -\frac{9u'}{y}, \qquad y'' = f_t + f_y f, \qquad u' = v[v + \rho(2x - 1)].$$

The exact target plateaus and slopes yield conservative whole-segment enclosures $|u| < 0.107$ and $|u'| < 0.041$. On the bootstrap strip $0.70 \le y \le 1.60$, these imply $|f| < 1.70$ and

$$|y''| \le \frac{9|u'|}{0.70} + \left(\rho + \frac{9|u|}{0.70^2}\right)\left(0.60\rho + \frac{9|u|}{0.70}\right) < 5.$$

A tighter value of this last bound is recomputed with outward-rounded rational arithmetic on each of the six target segments. On a fixed grid of $h_0 = 1/2000$, a scalar comparison principle is combined with the one-step Taylor enclosure

$$y(t + h_0) \in y(t) + h_0 f(t, y(t)) + [-B_j h_0^2/2, B_j h_0^2/2],$$

where B_j is the certified segment remainder bound. If the beginning state belongs to $[l, u]$, comparison of scalar ODE solutions gives the next lower bound from $l + h_0 f(t, l) - B_j h_0^2/2$ and the upper bound from $u + h_0 f(t, u) + B_j h_0^2/2$, each rounded outwards. A local $|f| < 1.70$ excursion bound encloses the solution between grid points and verifies, inductively, that it stays in the bootstrap strip. The bound is reinitialized at each piecewise-linear slope discontinuity without taking an artificial step across it.

There are exactly $80,000$ certified grid steps. For the path in Proposition 7.6, the machine-checkable output gives $0.7534859 < y(t) < 1.223386$ over the first day and $1.0301664 < y(t) < 1.406405$ over the second. It also encloses $1.2219388 < y(20) < 1.2225356$ and $1.4047874 < y(40) < 1.4055543$, establishing the moving 25% second-session limits and a terminal interval containing the high-precision reference value 1.405160822. For the nine target assets, the target coordinates follow exact line segments, with their daily lows attained at the prescribed closes and their daily highs below the permitted ceilings. Within-step interval ratios enclose both $|u|/x$ and $9|u|/y$ and give the gross-rate bound in (7.19). The same enclosures, with outward-rounded time sums, bound individual gross share sales and executed-dollar turnover, as claimed.

Finally, the terminal mean loss is $-\log[(0.9409)^9 y(40)]/10$. The assertion that it exceeds 0.020 is checked *without* relying on a floating-point logarithm: finite rational Taylor series provide lower and upper bounds on $e^{0.2}$ and verify $e^{0.2}(0.9409)^9 \sup y(40) < 1$. A second rational check at $e^{0.21}$ confirms the mean loss remains below 0.021. These assertions certify feasibility of the stated piecewise-linear path and its compensator ODE.

Price trajectories and gross executed-share rates. The following plots display the reconstructed price and executed-share-rate paths together with their certified envelope bounds. Figure 5 displays the price paths of the certified construction.

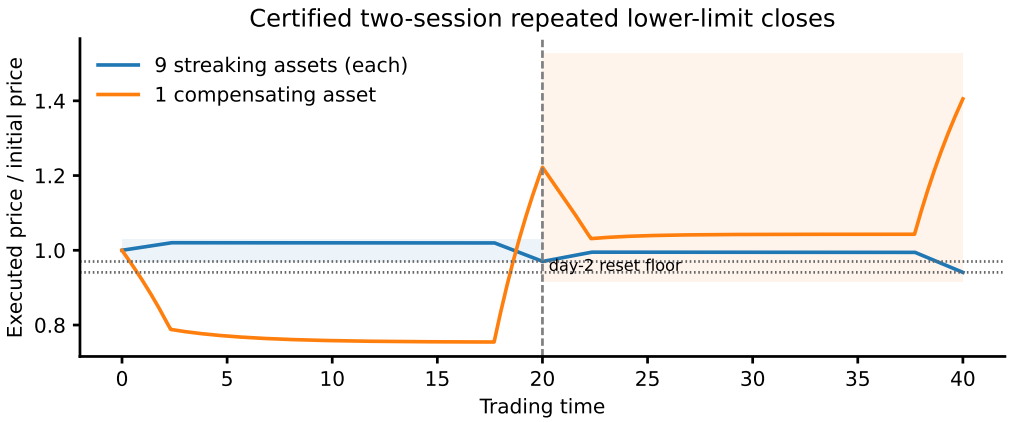


Figure 5. ***An interval-certified two-session path.*** *Continuous target and compensator prices for the construction in Proposition 7.6. The nine targets close at the reset lower limits; the compensator stays inside its 25% daily limits. Curves are illustrative numerical reconstructions; admissibility is independently certified by the outward-rounded arithmetic described in Appendix G.*

Figure 6 displays the corresponding gross execution rates.

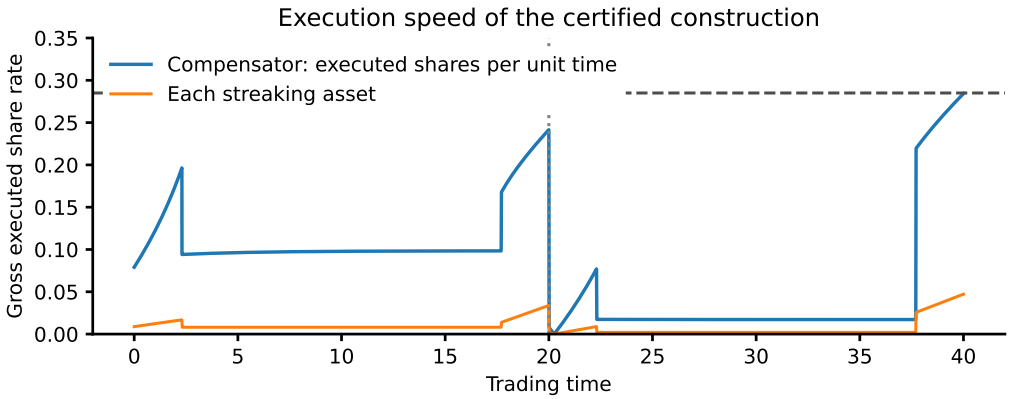


Figure 6. ***Gross execution rates in the certified path.*** *The largest computed gross share-rate is approximately* 0.28422*; rigorous interval rounding certifies a bound strictly below* 0.285*, without instantaneous pulses. The cap is per asset, before netting buys and sells.*

Sensitivity to cross-sectional composition. The ten-asset certificate deliberately reduces nine identical target states to one trajectory. Holding that *same* $x(t)$ and exact target-dollar closure fixed, a numerical sensitivity check repartitions ten assets into k identical targets and $10-k$ identical compensators. Their common compensator price obeys $\dot{y} = -0.4(y-1) - [k/(10-k)]u(t)/y$. Table 6 shows why the number of receiving assets matters: the specified control achieves the fixed 0.020 mean-loss target only for $k = 9$ among the cases tested. Changing k also changes the coverage fraction $k/10$; it is not a test at fixed $\alpha = 0.9$. For $k < 9$, the rows are high-accuracy ODE diagnostics with the control held fixed. The $k = 9$ witness depends on a wide compensator band and a receiving-capacity burden proportional to

$k/(10-k)$. For $k=9$, the feasibility bounds of Proposition 7.6 are interval-certified. Similarly, for the fixed $k=9$ path, the certificate proves compliance at a symmetric compensator band of $\pm 24.7\%$ and proves violation at $\pm 24\%$ (the price interval at $t=17.7$ lies strictly below 0.76). Changing the band does not rerun or optimize the control.

Table 6

Fixed-control sensitivity to cross-sectional composition. The tabulated prices, mean log losses, and band widths are high-precision numerical reconstructions. For $k=9$, feasibility under the stated constraints is additionally established by the interval certificate; Section 7.7 gives the certified bounds. The reported band width refers to the fixed path, without optimizing the control.

Targets k	Compensators	$y(40)$	Mean log loss	Computed band for fixed path
6	4	1.08112	0.00535	4.28%
7	3	1.12302	0.00784	6.55%
8	2	1.20141	0.01204	10.88%
9	1	1.40516	0.02081	24.55%

Conditional terminal threshold in the ten-asset case. For the interval-certified $N=10$, $k=9$, $q=0.97$, $D=2$ construction of Proposition 7.6, $D\lambda \approx 0.060918$. The compensator closes at $Y/F \approx 1.405161$ in the high-accuracy reconstruction, yielding $\mathcal{C}_{0.9} \approx 0.020811$. The outward-rounded certificate gives $1.4047874 < Y/F < 1.4055543$, which rigorously implies $\mathcal{C}_{0.9} > 0.020$ for the specified trajectory. The threshold $\delta = 0.020$ requires $Y/F \le \exp(9D\lambda - 10(0.020)) \approx 1.416609$, which the certified path satisfies. Thus the receiving asset's terminal price determines whether the same broad lower-limit streak qualifies as a systemic loss event.

Figure 7 illustrates the distinction between repeated closing-limit losses and the systemic crisis criterion.

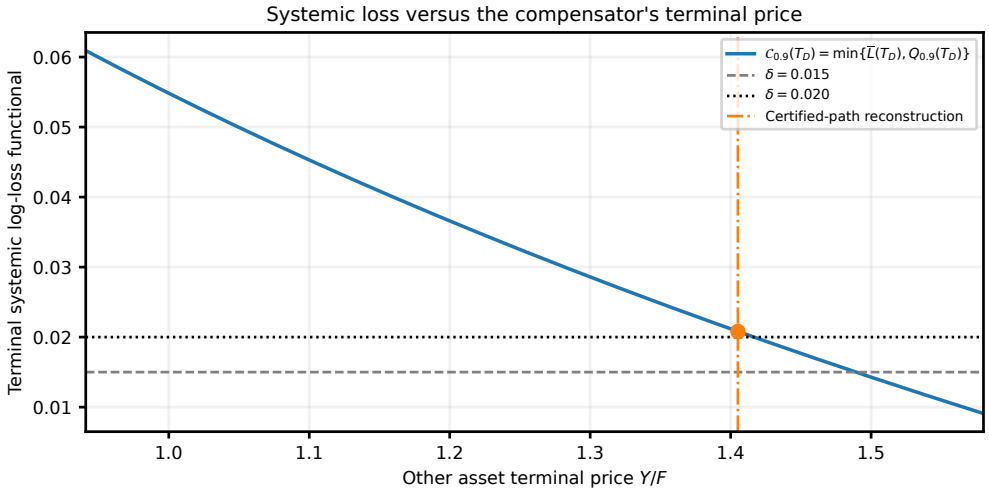


Figure 7. ***Systemic loss and the compensating asset's terminal price.*** *Conditional terminal severity for $N = 10$, nine assets closing twice at a 3% daily lower limit, versus the remaining asset's terminal price. The dotted marker denotes the constructed dollar-closed path; the reference lines show the log-loss thresholds $\delta = 0.015$ and $\delta = 0.020$. The curve gives the conditional algebraic relation between the terminal price and systemic loss.*

Appendix H. Sublinear power-law extension of uniform-budget severity. We prove Theorem 4.4 for fixed $0 \le p < 1$. Write $\nu = 2 - p \in (1, 2]$ and $m = N - 1$. The proof generalizes the radial construction of Appendix C; in particular, all resource bounds must remain uniform as the target price ε tends to zero.

Generalized radial trajectories. For $g_p(S) = \eta(S/F)^p$, the executed-dollar ledger is

$$X_i(T) = \frac{F^p}{\eta\nu}\big(S_i(T)^\nu - F^\nu\big) + \frac{\rho F^p}{\eta}\int_0^T S_i(t)^{\nu-1}\big(S_i(t) - F\big)\,\mathrm{d}t. \tag{H.1}$$

Each flow-completed paired-dollar pulse conserves $\sum_i S_i^\nu$. Define the generalized radius by $R^\nu = \sum_i S_i^\nu$. For any positive direction $\boldsymbol{u} = (u_1, \dots, u_N)$ satisfying $\sum_i u_i^\nu = 1$, set $H = \sum_i u_i^{\nu-1}$ and $S_i(t) = u_i R(t)$. The aggregate price equation is solved by

$$\begin{aligned} R(t) &= F\big[H + (N^{1/\nu} - H)e^{-\rho t}\big], \\ \dot{X}_i(t) &= \frac{\rho F^{p+1}}{\eta} R(t)^{\nu-1} u_i^{\nu-1}(Hu_i - 1). \end{aligned} \tag{H.2}$$

Indeed, $\sum_i \dot{X}_i = 0$ because $H\sum_i u_i^\nu = \sum_i u_i^{\nu-1}$. Hölder's inequality gives $1 \le H \le N^{1/\nu}$; hence $F \le R(t) \le FN^{1/\nu}$. All stage flows are admissible paired transfers through the complete bidirectional graph in Assumption 2.1.

A uniformly positive compensating direction. Fix $\bar{b} = 1/5$. Choose $c \in (0, 1/10)$ sufficiently small that $(m-1)c^{\nu-1} < 1/4$ and $(m-1)c^\nu < 1/4$. For $b \in [0, \bar{b}]$, define

$$a = (1 - b^\nu - (m-1)c^\nu)^{1/\nu}, \quad H_b = a^{\nu-1} + (m-1)c^{\nu-1} + b^{\nu-1}.$$

Then $a > \max\{b, c\}$ and $H_b < 9/4$, so $bH_b < 1$ and $cH_b < 1$. Use the m directions $\boldsymbol{u}_j = (c, \dots, \underset{j}{a}, \dots, c, b)$: the compensator has weight b throughout, while each of the m targets has weight a during exactly one stage. Every direction has the same H_b and therefore the same radius $R_b(t)$ in (H.2).

At time 0, a fixed-radius positive-price pulse takes $F\mathbf{1}$ to $R_b(0)\boldsymbol{u}_1$. Between stages we switch directions by fixed-radius pulses. At T, a final such pulse sets

$$S_1(T) = \cdots = S_m(T) = \varepsilon, \qquad S_N(T) = \big(R_b(T)^\nu - m\varepsilon^\nu\big)^{1/\nu}.$$

All pulse endpoints and within-pulse prices are positive when $\varepsilon > 0$ is sufficiently small. Monotone interpolation in the coordinates S_i^ν and the complete paired-dollar graph implement these pulses with nonnegative edge masses.

By (H.1), the compensator's terminal executed-dollar balance equals

$$\mathcal{F}_\varepsilon(b) = \frac{F^p}{\eta\nu}\big(R_b(T)^\nu - m\varepsilon^\nu - F^\nu\big) + \frac{\rho F^p}{\eta}\int_0^T (bR_b(t))^{\nu-1}(bR_b(t) - F)\,\mathrm{d}t.$$

For $b = 0$ use only the continuous extension: the integrand tends to zero, because $\nu - 1 > 0$, and $\mathcal{F}_0(0) > 0$, since $R_0(T) > F$ for every finite T. In contrast, at $b = \bar{b}$ the stage compensator price converges as $t \to \infty$ to $F\bar{b}H_{\bar{b}} < F$. Its negative recovery contribution therefore grows linearly in T, while the endpoint term remains bounded. Choose a finite T large enough that $\mathcal{F}_0(\bar{b}) < 0$. By continuity, choose $b_- > 0$ with $\mathcal{F}_0(b_-) > 0$. The formula

$$\mathcal{F}_\varepsilon(b) = \mathcal{F}_0(b) - \frac{mF^p}{\eta\nu}\varepsilon^\nu$$

shows that, for every sufficiently small $\varepsilon > 0$, some $b_\varepsilon \in (b_-, \bar{b})$ gives $\mathcal{F}_\varepsilon(b_\varepsilon) = 0$. The horizon T, the direction parameter c, and the positive lower bound b_- are fixed independently of ε.

Simultaneous closure of the target ledgers. For $b = b_\varepsilon$, the difference between a target's high-state and low-state recovery integrands is

$$W(t) = R_b(t)^{\nu-1}\Big[(a^\nu - c^\nu)R_b(t) - F(a^{\nu-1} - c^{\nu-1})\Big].$$

This is strictly positive: $R_b(t) \geq FH_b$, $cH_b < 1$, and

$$aH_b - 1 = (m-1)c^{\nu-1}(a - c) + b^{\nu-1}(a - b) > 0.$$

Indeed, $H_b(a^\nu - c^\nu) - (a^{\nu-1} - c^{\nu-1}) = a^{\nu-1}(aH_b - 1) + c^{\nu-1}(1 - cH_b) > 0$. Partition $[0, T]$ into m consecutive intervals, each having the same positive integral of W. Let target j have high direction weight a on interval j and low weight c elsewhere. All target recovery integrals in (H.1) are then equal. Their initial and terminal prices are also equal, so $X_1(T) = \cdots = X_m(T)$. Paired-dollar conservation gives $\sum_i X_i(T) = 0$, and the compensator has $X_N(T) = 0$; hence $X_i(T) = 0$ for every asset.

Turnover and share inventory bounds. There are $m+1=N$ fixed-radius pulses. During each pulse the coordinates S_i^ν vary monotonically, and $\mathrm{d}X_i = (F^p/(\eta\nu))\,\mathrm{d}(S_i^\nu)$. Since $0 < S_i \le FN^{1/\nu}$ throughout the constructed path, each pulse costs at most $N^2F^2/(2\eta\nu)$ in paired-dollar turnover. During controlled stages, (H.2) and $\sum_i u_i^\nu = 1$ give

$$\frac{1}{2}\sum_i |\dot{X}_i| \le \frac{\rho F^{p+1}}{\eta} R^{\nu-1} H_b \le \frac{\rho N F^2}{\eta}.$$

The complete graph realizes this turnover by routing negative dollar flows to positive ones. Thus a uniform admissible bound is

$$M(\gamma) \le M_0 := \frac{N^3F^2}{2\eta\nu} + \frac{\rho N F^2 T}{\eta}.$$

For a pulse, the share increment satisfies $\mathrm{d}\theta_i = (F^p/\eta)S_i^{-p}\,\mathrm{d}S_i$. Because $\nu - 1 = 1 - p > 0$, its cumulative share sale is at most $FN^{(\nu-1)/\nu}/[\eta(\nu-1)]$ per asset. Put $d_* = \min\{c, b_-\} > 0$. On every controlled stage, $u_i \ge d_*$ and

$$|\dot{\theta}_i| = \frac{\rho F^{p+1}}{\eta} R^{-p} u_i^{-p} |Hu_i - 1| \le \frac{\rho F}{\eta} d_*^{-p}(1 + N^{1/\nu}).$$

Consequently the total share sales of each asset satisfy

$$H_i^-(\gamma) \le H_0 := \frac{NF}{\eta(\nu-1)} N^{(\nu-1)/\nu} + \frac{\rho F T}{\eta} d_*^{-p}(1 + N^{1/\nu}). \tag{H.3}$$

These finite constants do not depend on ε. Initial inventory of H_0 shares per asset therefore prevents short sales.

Finally, the m targets have log loss $\log(F/\varepsilon) \to \infty$, while the compensator's positive terminal price is bounded above by $FN^{1/\nu}$. The average log loss and the kth-largest loss both diverge for $k \le m$. This proves $\mathcal{C}_\alpha(T) > \delta$ for every finite $\delta > 0$ with the same T, M_0, H_0. The blow-up of $\Psi_p(0^+)$ as $p \uparrow 1$ precludes a corresponding claim of resource constants uniform up to $p = 1$.

Appendix I. Reproducing the uniform-band loss–breadth witnesses. Table 7 compares six additional two-session constant-impact paths. Their exact rational piecewise-constant dollar-rate input is defined in `verification/loss_breadth_certified.py`; a separate discussion of assumptions, evidence levels and comparisons appears in `docs/LOSS_BREADTH_CERTIFICATES.md`. For k identically traded target assets and $m = 10 - k$ compensators, each representative pair solves

$$\dot{x} = -\frac{2}{5}(x-1) + \frac{u(t)}{x}, \qquad \dot{y} = -\frac{2}{5}(y-1) - \frac{k}{m}\frac{u(t)}{y}.$$

Dollar matching holds identically and each rational schedule has $\int_0^{40} u(t)\,\mathrm{d}t = 0$, giving zero terminal executed dollars per asset. The verification uses outward-rounded integer Taylor enclosures, scale 10^{24} and steps no larger than $1/4000$, bounded first and second derivatives

Table 7

Systemic loss–breadth comparison for specified positive-price dollar-closed controls. *All new cases use uniform two-session ±25% rolling bands, unlike the original 3%-target/25%-compensator witness. Target loss denotes an ordinary price decline in at least k assets; index loss denotes the decline of the equal-weight geometric price index. Prefixes* `P` *and* `R` *identify, respectively, two-window and three-window piecewise-constant trading templates; the labels indicate coverage percentage and nominal target loss, and* `M5` *denotes the relaxed $M < 5$ budget. Loss thresholds are conservative strict lower bounds, whereas the M and gross-rate columns give certified upper bounds.*

Witness	k	Target loss	Index loss	M upper	Max. gross rate	M cap
Original	9	5.91%	$> 1.98\%$	2.984	0.284500	3
P50-20	5	$> 22.29\%$	$> 4.0\%$	2.600	0.216130	3
P60-20	6	$> 20.90\%$	$> 5.0\%$	2.880	0.257224	3
P70-10	7	$> 12.60\%$	$> 3.0\%$	2.660	0.164249	3
R80-10	8	$> 10.80\%$	$> 3.0\%$	2.851	0.272696	3
R70-15.5	7	$> 15.50\%$	$> 4.0\%$	2.940	0.277592	3
R70-20-M5	7	$> 20.20\%$	$> 6.0\%$	4.600	0.262743	5

on a verified positive-price strip, and the monotonicity of each scalar autonomous segment. It verifies every segment's price range and both rolling session bands, upper bounds on gross share sales and share rate, and paired-dollar turnover. A rational endpoint product test, $x_{\text{upper}}^k y_{\text{upper}}^m < q^{10}$, certifies the geometric-index price decline $> 1 - q$ without a floating-point logarithm.

These interval computations establish feasibility of the specified controls at the budgets and rates reported in Table 7.

Fixed-budget and relaxed-budget comparison. The primary loss–breadth certificate family is summarized below. Its target losses refer to ordinary individual price declines, while index losses refer to the equal-weight geometric price index.

Additional loss–breadth witnesses. The exact-rational verification archive `verification/loss_breadth/` extends the six new controls in Table 7 to twelve. Table 8 records the remaining six under the same uniform two-session ±25% price bands, benchmark parameters and gross-rate and share-inventory restrictions. The controls are prescribed feasible schedules, with individual executed-dollar ledgers closing exactly.

Table 8

Further specified loss–breadth controls. *Target and geometric-index losses are strict conservative ordinary percentage-decline thresholds; dollar turnover and gross share rate are reported as conservative upper bounds. Here k counts the number of terminally qualifying assets: the full-coverage controls have two five-asset groups, both declining at the horizon. The final two rows use higher turnover caps, distinct from the $M < 3$ comparisons.*

Witness	k	Target loss	Index loss	M upper	Max. gross rate	M cap
R70-16.4	7	$> 16.40\%$	$> 4.0\%$	2.971	0.282729	3
R90-8	9	$> 8.00\%$	$> 3.0\%$	2.971	0.282957	3
F100-0.65	10	$> 0.65\%$	$> 0.65\%$	2.854	0.259746	3
F100-0.75	10	$> 0.75\%$	$> 0.75\%$	2.970	0.281922	3
R70-20-M4	7	$> 20.00\%$	$> 6.0\%$	3.961	0.282743	4
F100-2-M7	10	$> 2.00\%$	$> 2.0\%$	6.732	0.281818	7

The complete twenty-four-cell fixed-$M < 3$ research matrix, strict interval bounds, and hashes of exact controls are available in `verification/loss_breadth/`. Its `OPEN` entries indicate unclassified cases under the available controls, while `OLD` marks thresholds already reached by the earlier, tighter target-band witness. These comparisons supplement the existence certificates without altering the analytical exclusion and resource bounds.

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